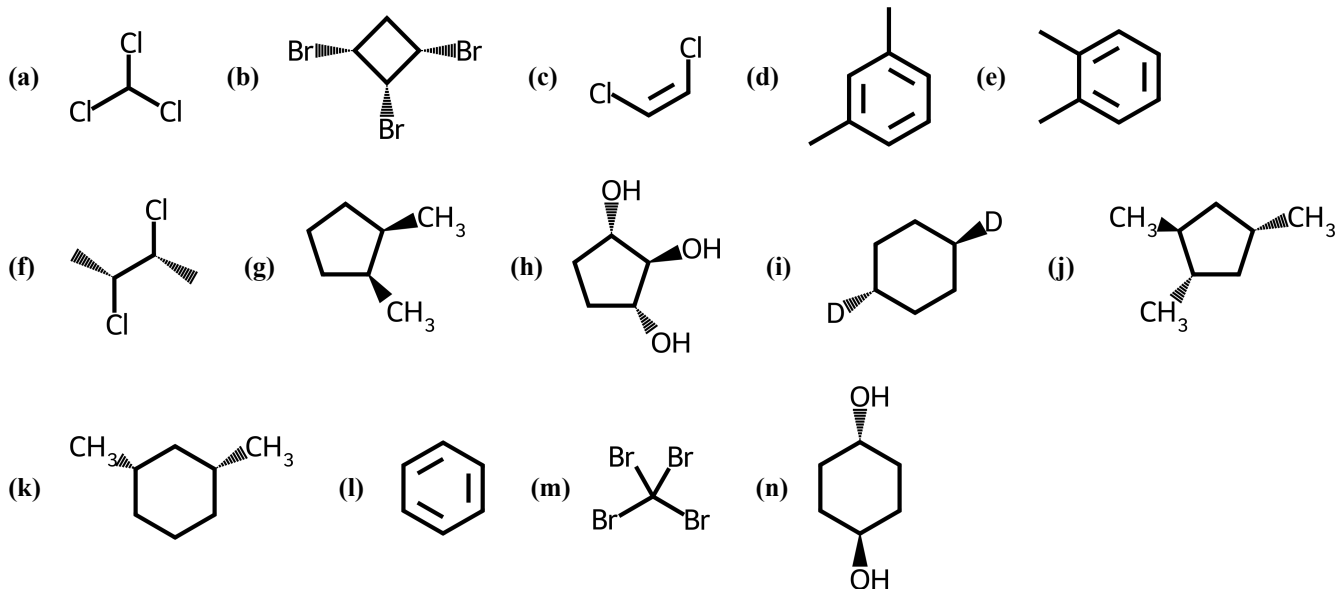


OPTICAL ISOMERISM

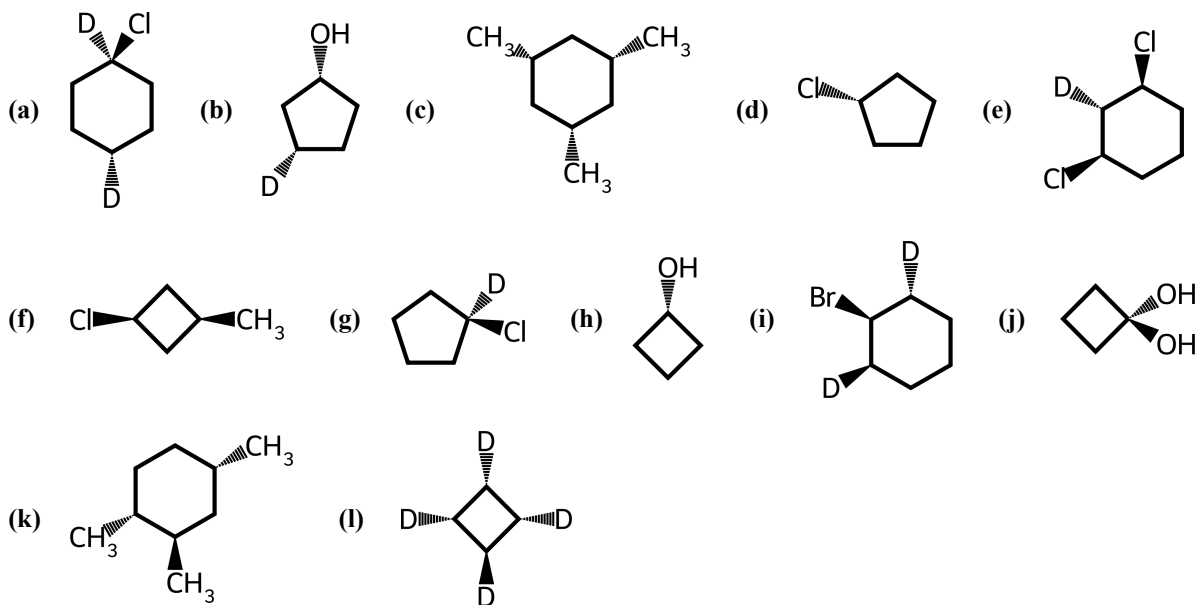
Part A — Elements of symmetry

Use idealised geometries: rings planar; a wedge = group above the plane, a dashed bond = below it; no drawn group = H.

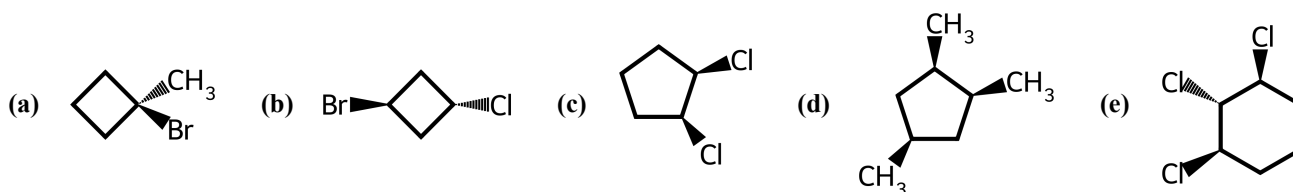
1. For each of the following molecules mark the elements of symmetry present among POS (plane of symmetry), COS (centre of symmetry) and AOS (axis of symmetry, a proper rotation axis C_n with $n \geq 2$). Find the total number of elements marked over all 14 molecules.

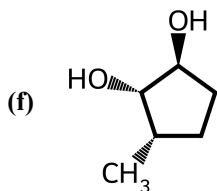


2. How many of the following are optically active (chiral)?

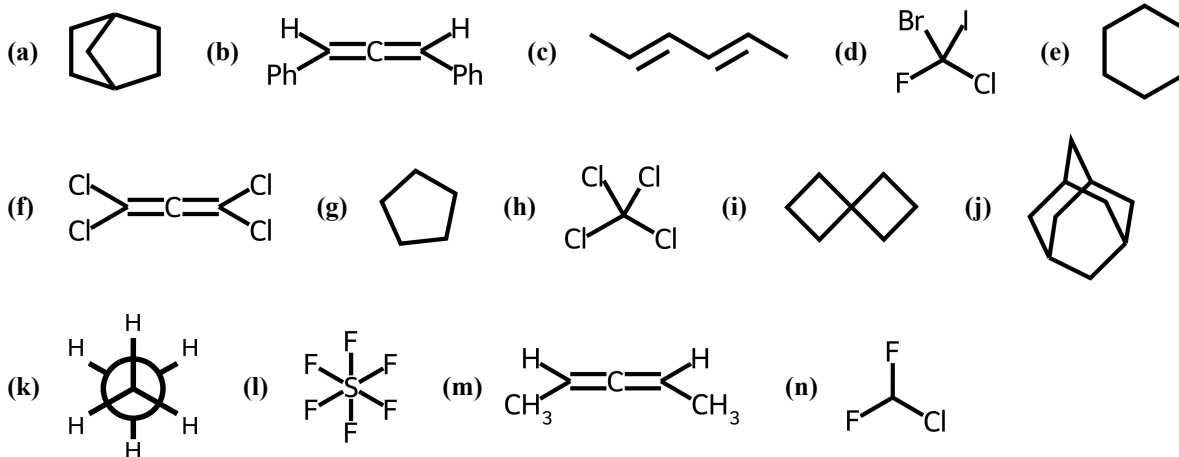


3. Find the total number of planes of symmetry present in the compounds (a)–(f) taken together.





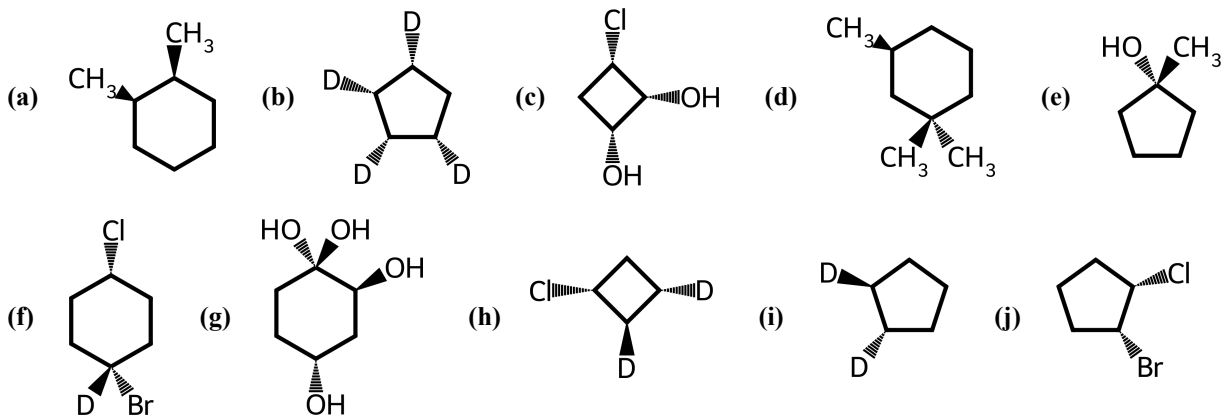
4. How many of the following possess an ALTERNATE axis of symmetry (a rotation–reflection axis S_n with $n \geq 3$)?

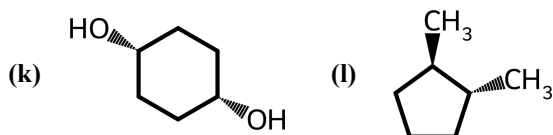


5. Read the following 14 statements on symmetry elements and find how many of them are correct.

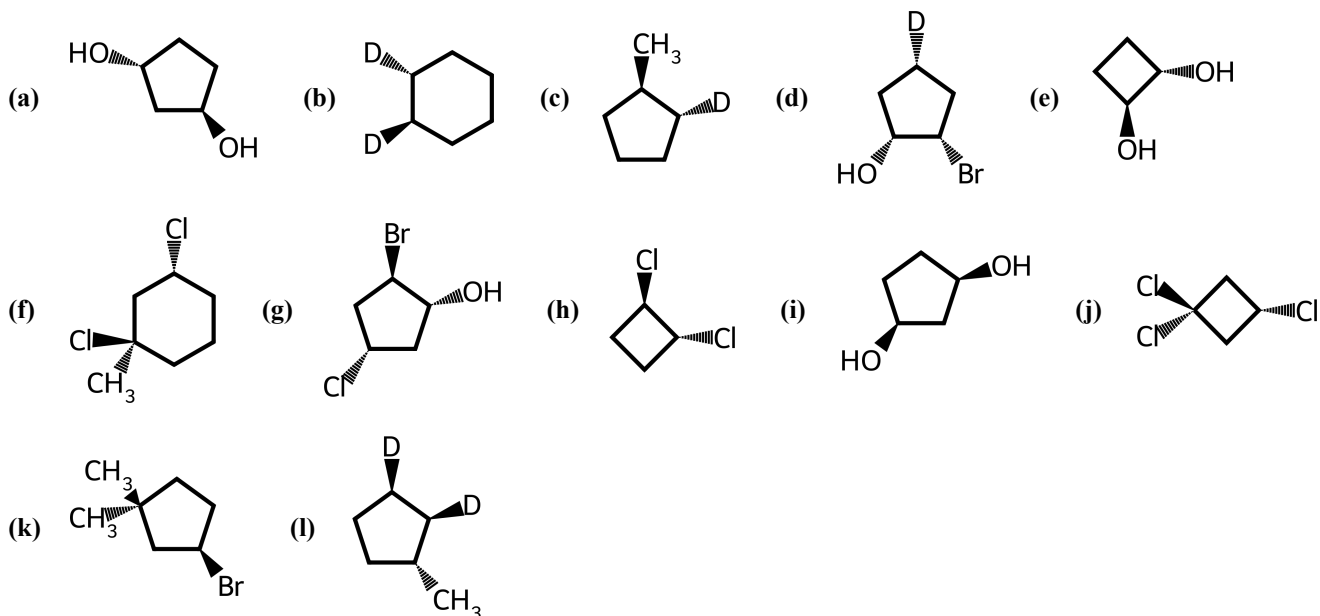
- (1) A molecule that possesses a plane of symmetry is achiral.
- (2) A molecule that possesses a centre of symmetry can be chiral.
- (3) (R,R)-Tartaric acid has only a C_2 axis and is chiral.
- (4) S_1 is equivalent to a plane of symmetry and S_2 to a centre of symmetry.
- (5) A molecule with an S_4 axis but neither a plane nor a centre of symmetry is achiral.
- (6) Methane has four C_3 axes, three C_2 axes, three S_6 axes and six planes of symmetry.
- (7) Benzene has a C_6 axis, six C_2 axes perpendicular to it, seven planes of symmetry and a centre of symmetry.
- (8) Planar cyclohexane has a centre of symmetry.
- (9) cis-1,3-Dimethylcyclohexane (planar ring) has a plane of symmetry.
- (10) trans-1,2-Dimethylcyclohexane (planar ring) has a plane of symmetry.
- (11) trans-1,4-Dimethylcyclohexane (planar ring) has a centre of symmetry.
- (12) cis-1,4-Dimethylcyclohexane (planar ring) has a centre of symmetry.
- (13) Staggered ethane has a centre of symmetry and an S_6 axis.
- (14) Propadiene (allene) has a centre of symmetry.

6. How many of the following are ASYMMETRIC (possess no element of symmetry at all)?

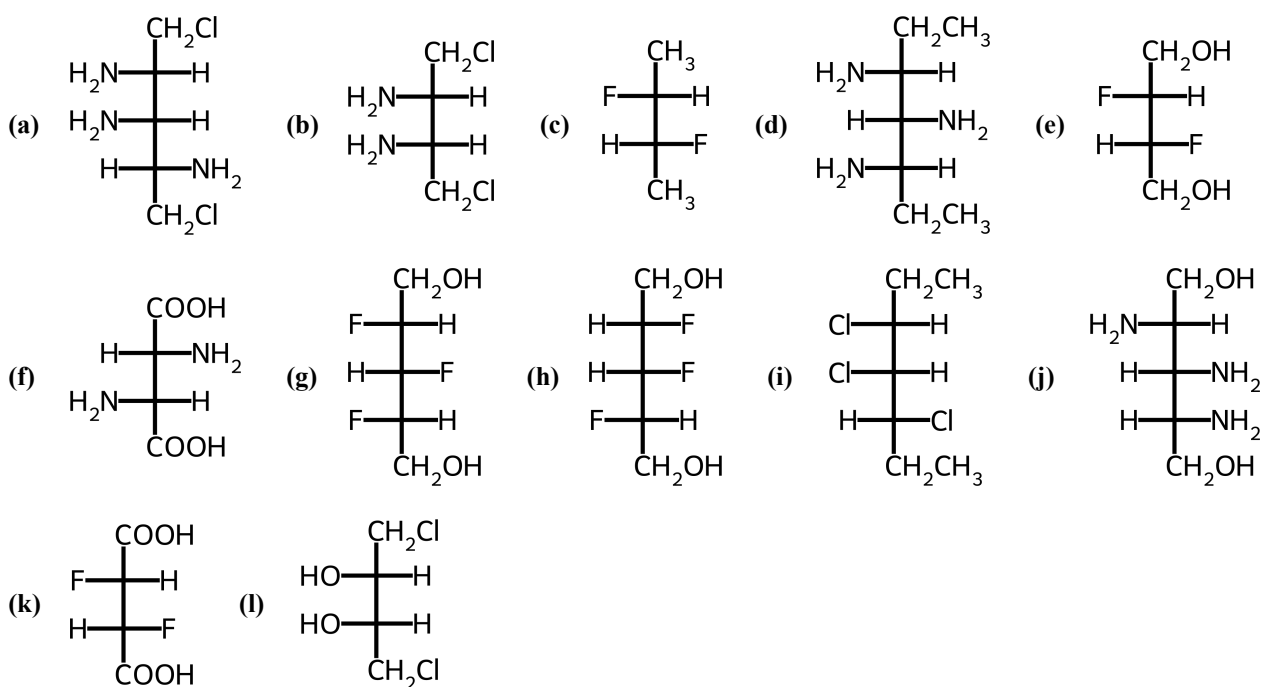




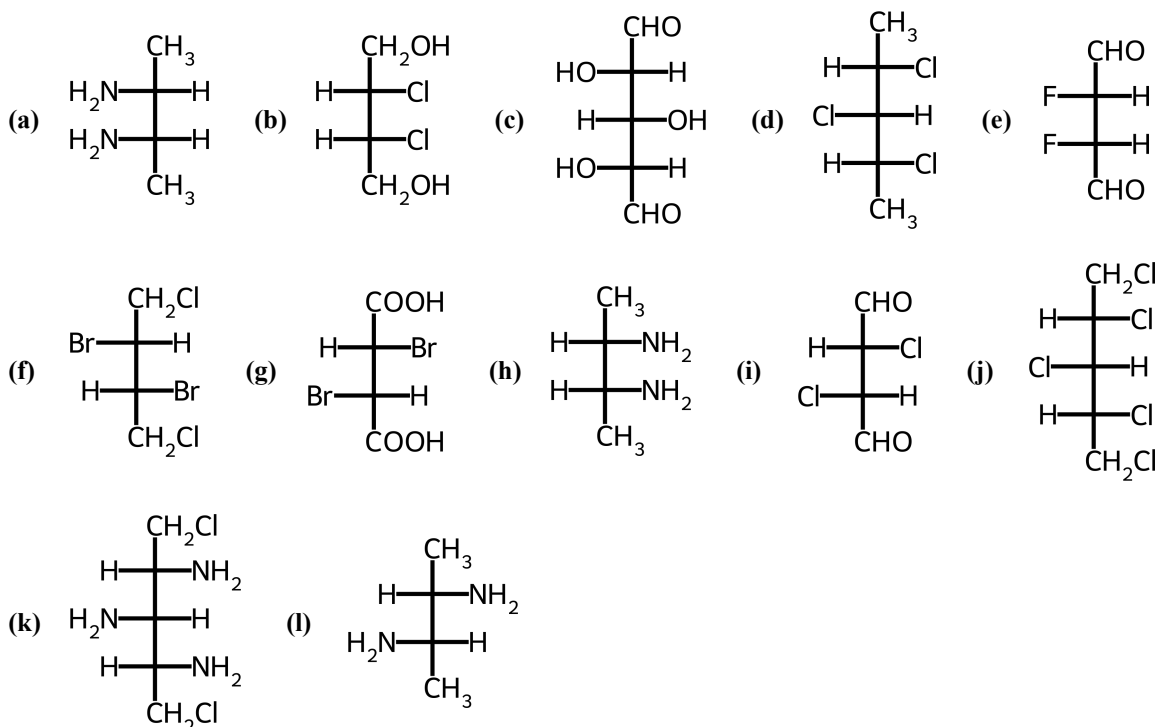
7. How many of the following are DISSYMMETRIC but not asymmetric (chiral, yet possessing a proper axis of symmetry)?



8. How many of the following Fischer projections (eclipsed conformation, as drawn) possess a PLANE of symmetry?



9. How many of the following Fischer projections (eclipsed conformation, as drawn) possess a C_2 axis of symmetry perpendicular to the plane of the paper through the middle of the chain?



10. Mark POS (plane), COS (centre), AOS (proper axis, C_n with $n \geq 2$) and AAOS (alternate axis, S_n with $n \geq 3$) for each of the following species. Find the total number of marks over all eight species.

- (a) envelope cyclopentane (b) anti-butane (c) trans-1,2-dimethylcyclohexane (diequatorial chair)
 (d) trans-decalin (e) staggered ethane (f) cis-1,2-dimethylcyclohexane (time-averaged)
 (g) gauche-1,2-dichloroethane (h) boat cyclohexane

11. How many of the following species are chiral (no plane, no centre and no alternate axis of symmetry)?

- (a) anti-1,2-dichloroethane (b) trans-1,4-dimethylcyclohexane (diequatorial chair)
 (c) (R,R)-tartaric acid (anti conformation) (d) chair cyclohexane (e) gauche-butane (f) adamantane
 (g) half-chair cyclohexene (h) syn-eclipsed butane (i) anti-butane (j) trans-decalin
 (k) cyclooctatetraene (tub) (l) trans-cyclooctene

12. How many of the following species possess a C_2 (or higher) axis of symmetry but NO plane of symmetry?

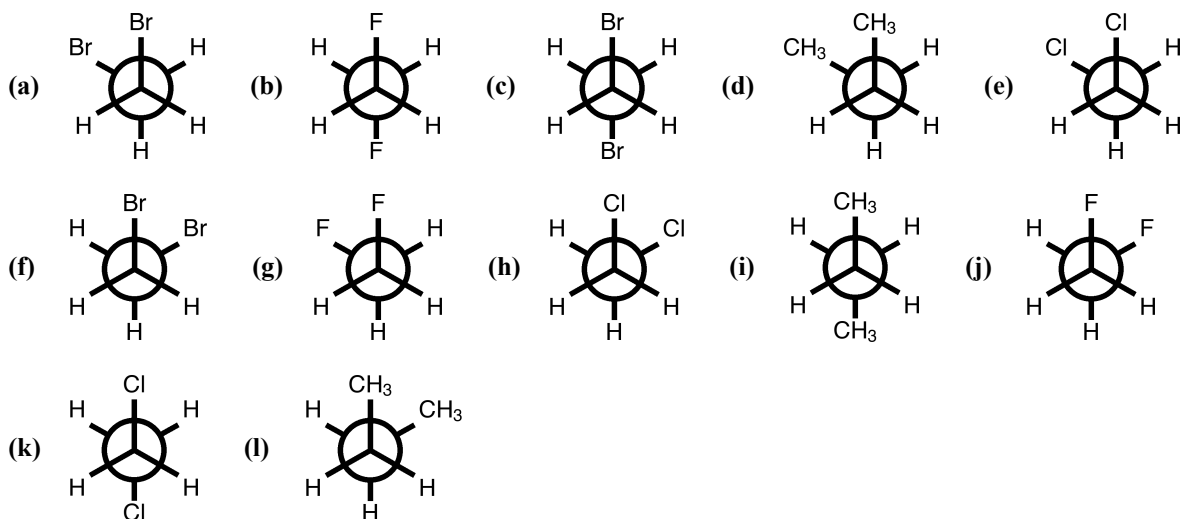
- (a) twistane (b) syn-eclipsed butane (c) boat cyclohexane (d) staggered ethane
 (e) gauche-1,2-dichloroethane (f) hexahelicene (g) cis-cyclooctene (crown/boat-chair)
 (h) chair cyclohexane (i) twist-boat cyclohexane (j) cis-1,3-dimethylcyclohexane (diequatorial chair)
 (k) half-chair cyclohexene (l) trans-cyclooctene

13. How many of the following species possess BOTH a plane and a centre of symmetry?

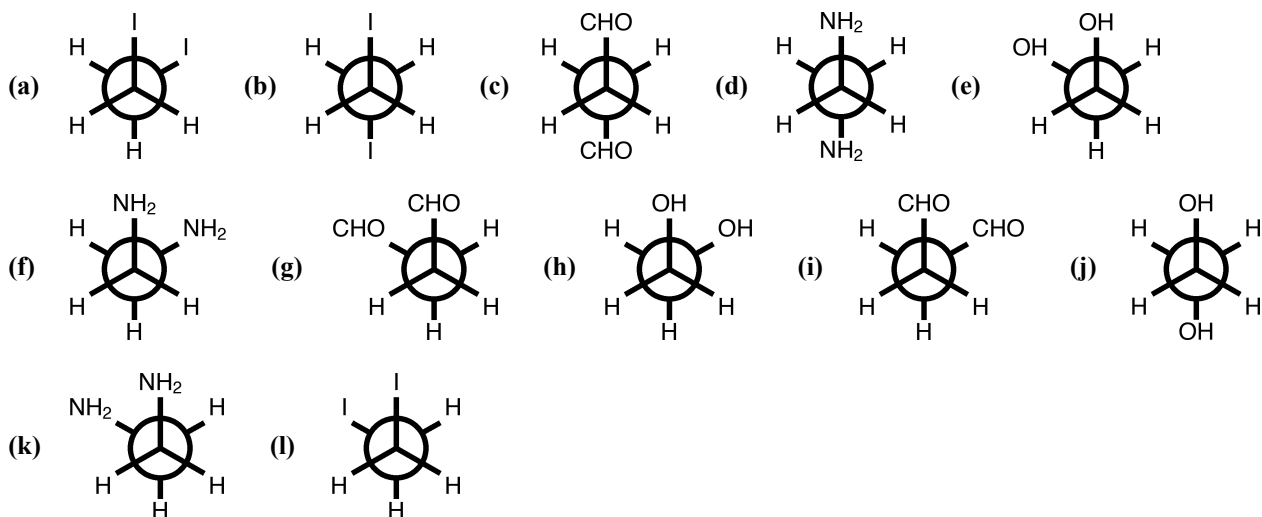
- (a) trans-1,4-dimethylcyclohexane (diequatorial chair) (b) staggered ethane (c) hexahelicene

- (d) trans-1,2-dimethylcyclohexane (diequatorial chair) (e) adamantane
- (f) cis-1,3-dimethylcyclohexane (diequatorial chair) (g) cis-1,4-dimethylcyclohexane (chair)
- (h) meso-tartaric acid (anti conformation) (i) gauche-butane (j) anti-1,2-dichloroethane
- (k) cyclooctatetraene (tub) (l) trans-cyclooctene

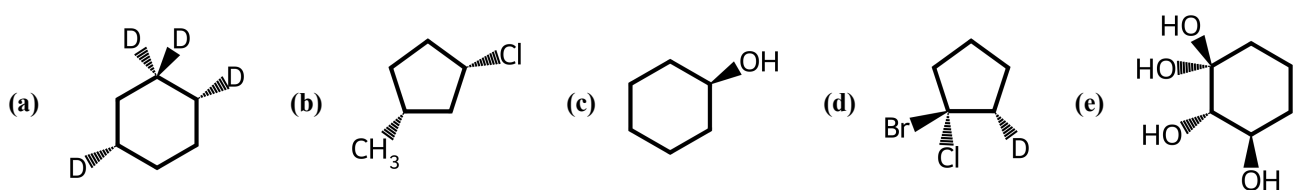
14. Each Newman projection below shows a staggered conformation of an $X-CH_2-CH_2-X$ molecule (the identical groups X are shown). How many of the conformations are chiral?

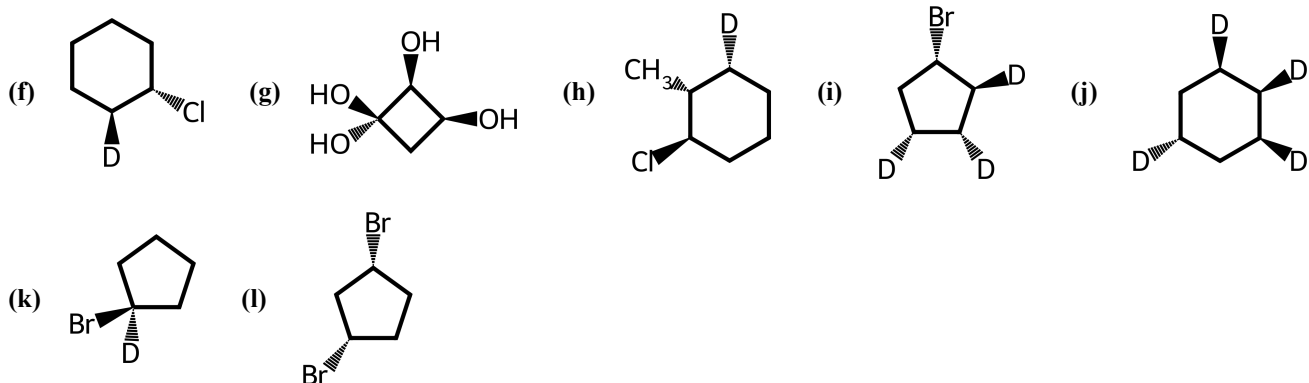


15. Each Newman projection below shows a staggered conformation of an $X-CH_2-CH_2-X$ molecule. How many of the conformations possess a CENTRE of symmetry?

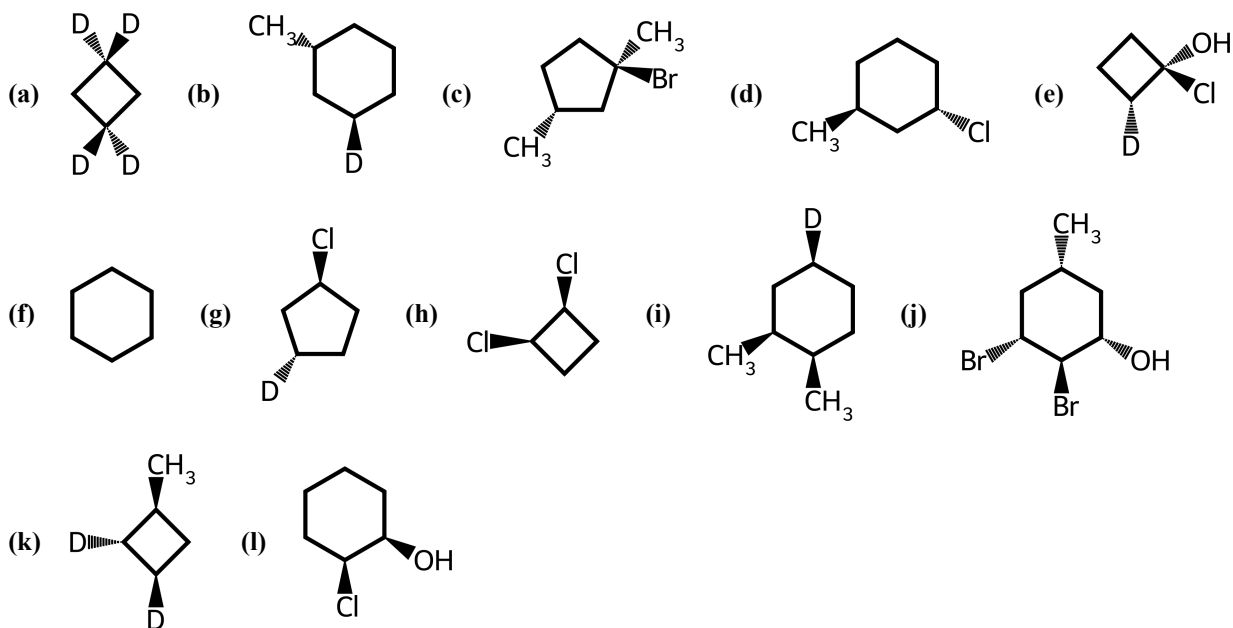


16. How many of the following have EXACTLY ONE plane of symmetry?

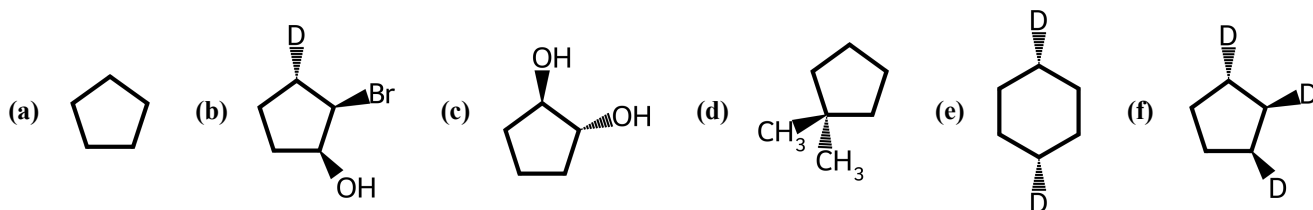




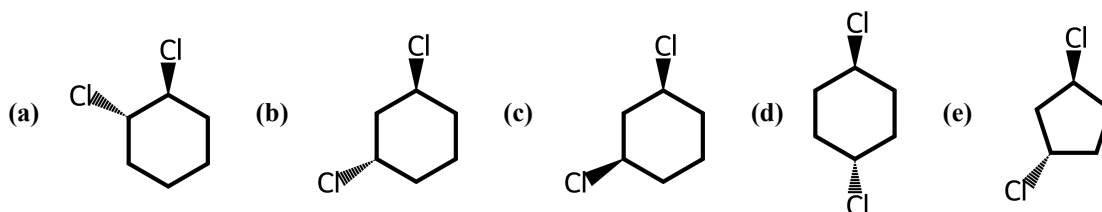
17. How many of the following have MORE THAN ONE axis of symmetry?

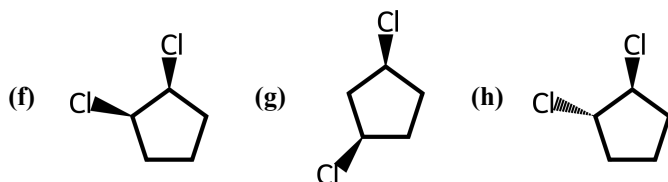


18. Find the total number of axes of symmetry (every C_n , $n \geq 2$, counted once per axis) present in the compounds (a)–(f) taken together.

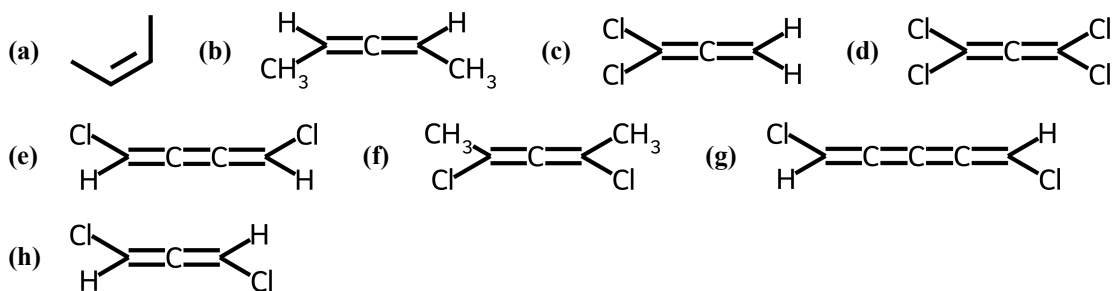


19. Mark POS (plane), COS (centre), AOS (axis, C_n with $n \geq 2$) and AAOS (alternate axis, S_n with $n \geq 3$) for each of the following compounds. Find the total number of marks.



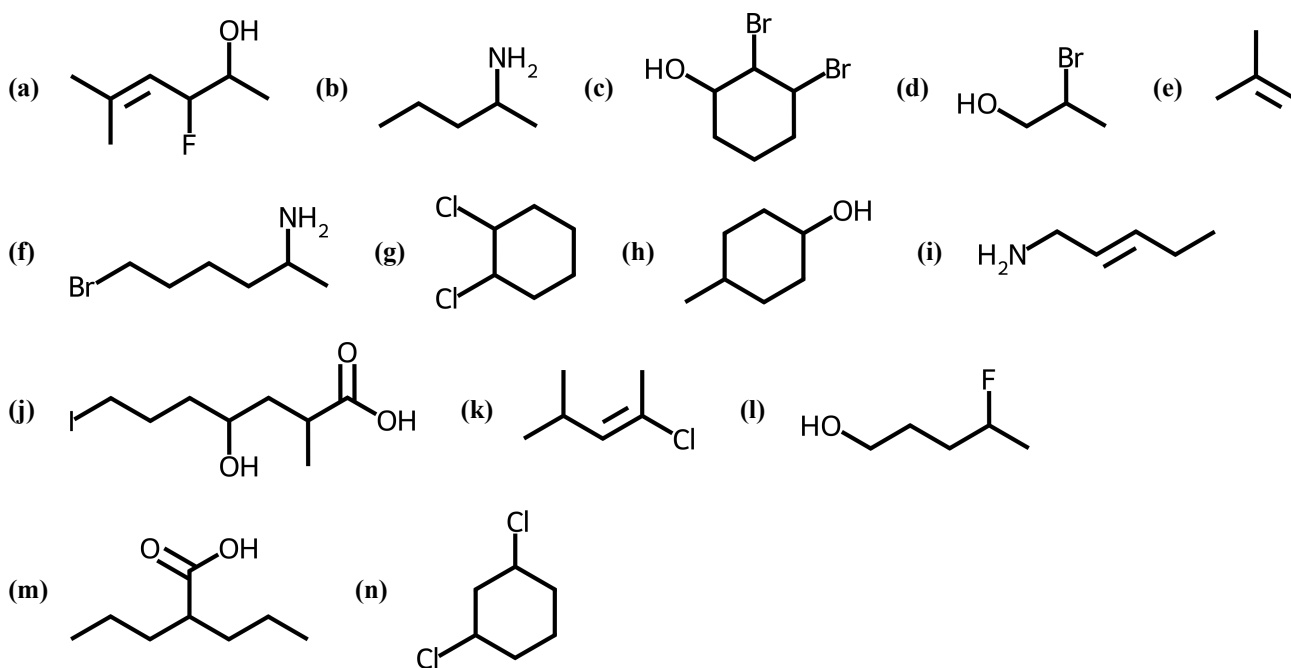


20. Mark POS (plane), COS (centre), AOS (axis, C_n with $n \geq 2$) and AAOS (alternate axis, S_n with $n \geq 3$) for each of the following alkenes and cumulenes (planar cis/trans isomers for an odd number of cumulated double bonds, axial chirality for an even number). Find the total number of marks.

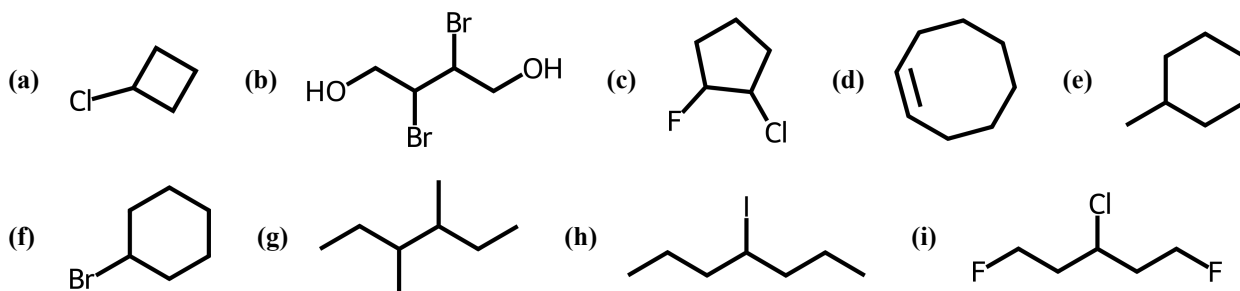


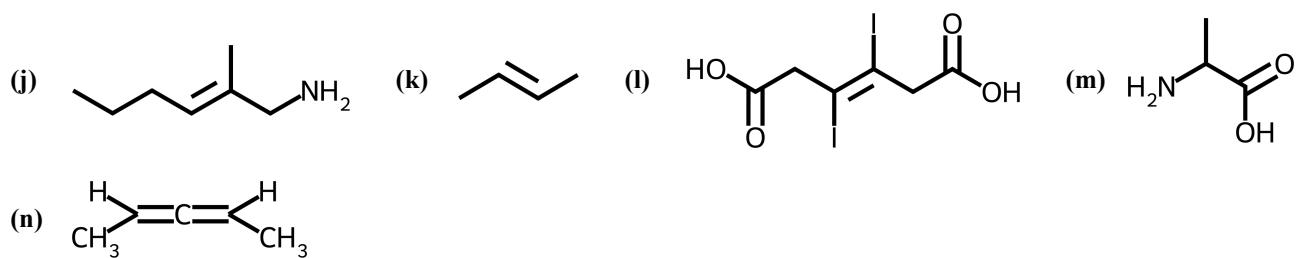
Part B — Identification of stereo-features (count the compounds)

21. How many of the following compounds can show optical isomerism?

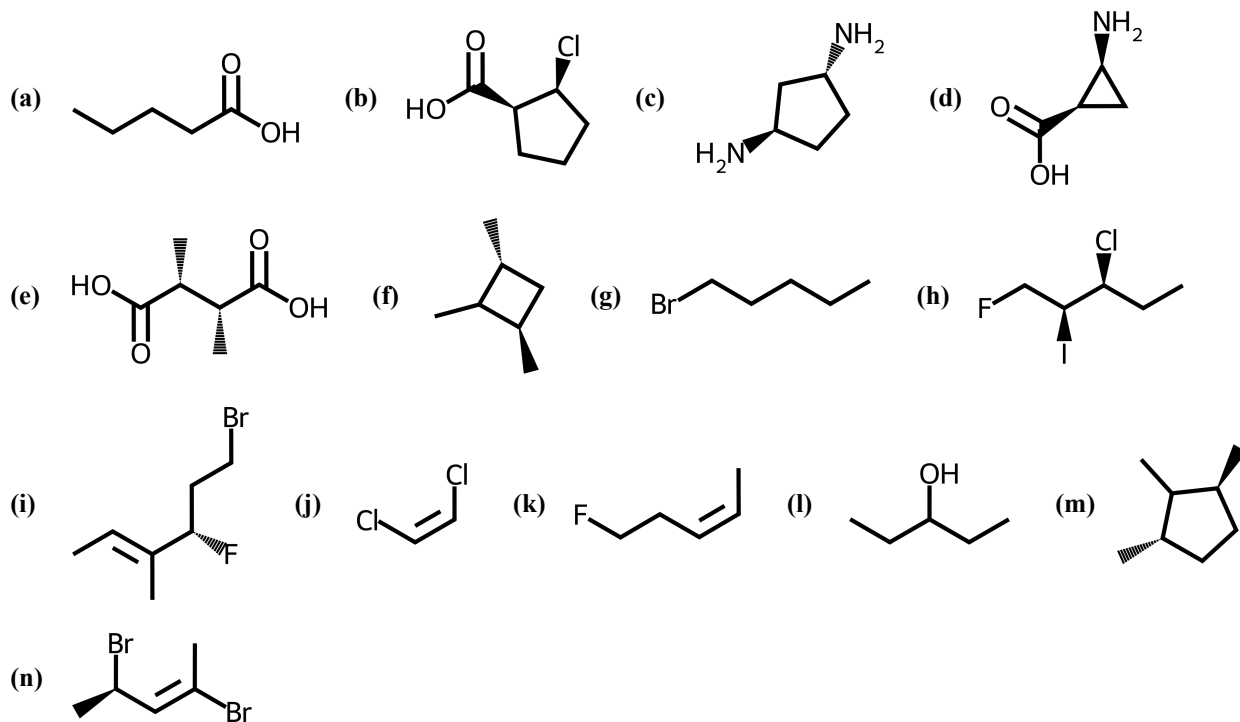


22. How many of the following compounds can show stereoisomerism (geometrical and/or optical)?

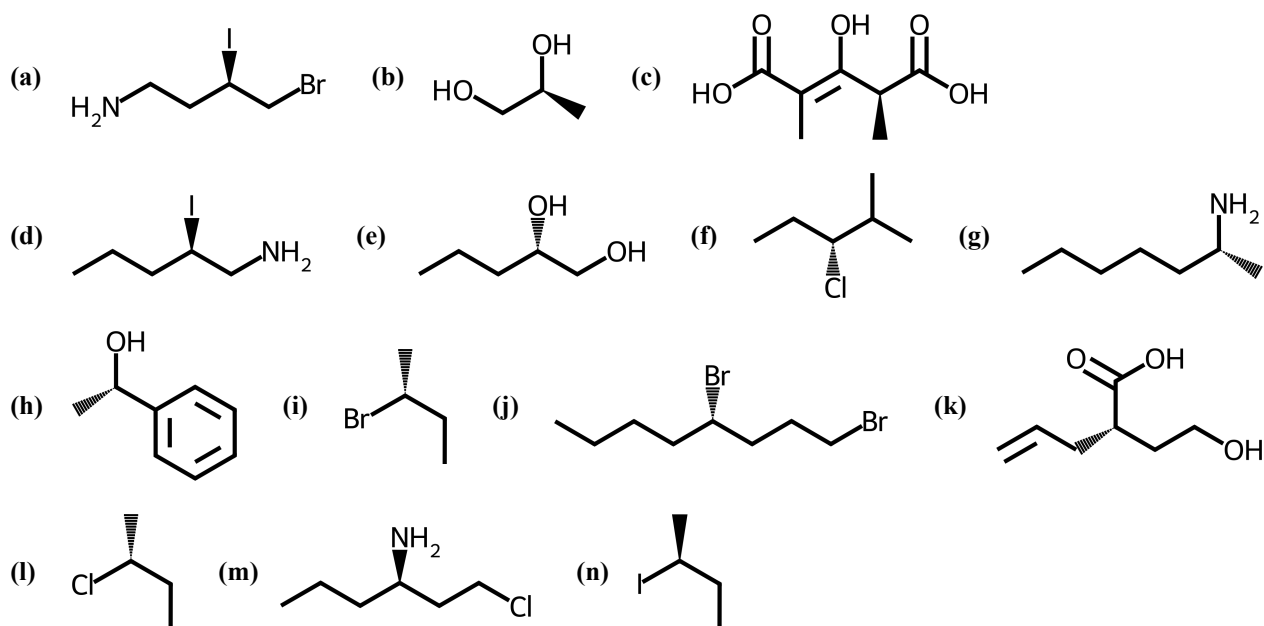




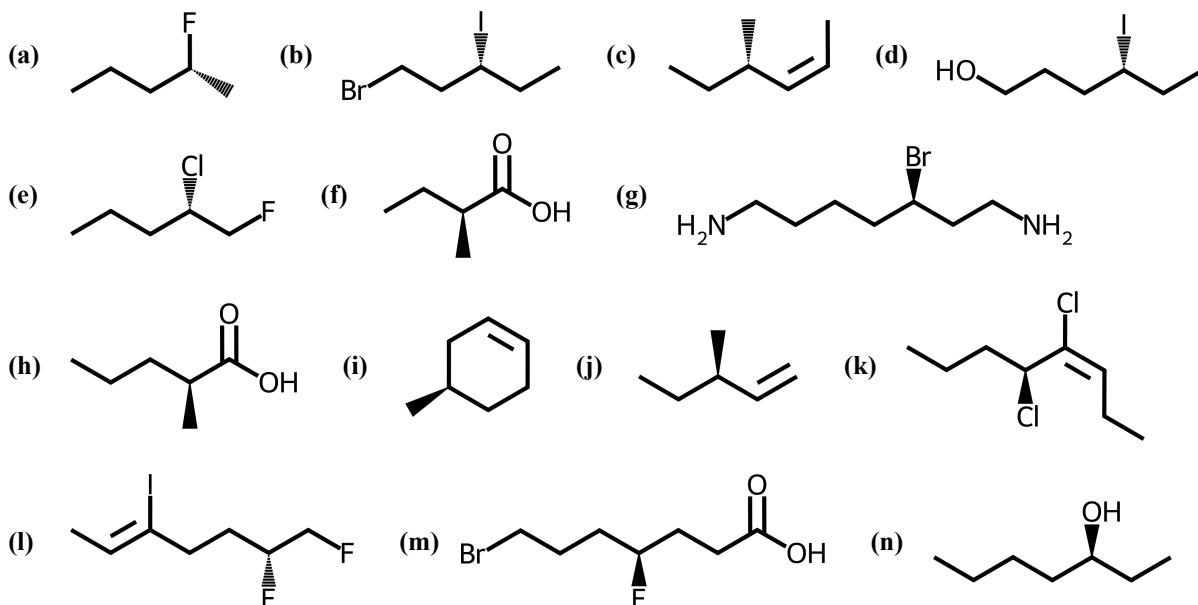
23. How many of the following molecules, as drawn, are optically active (chiral and a single pure stereoisomer)?



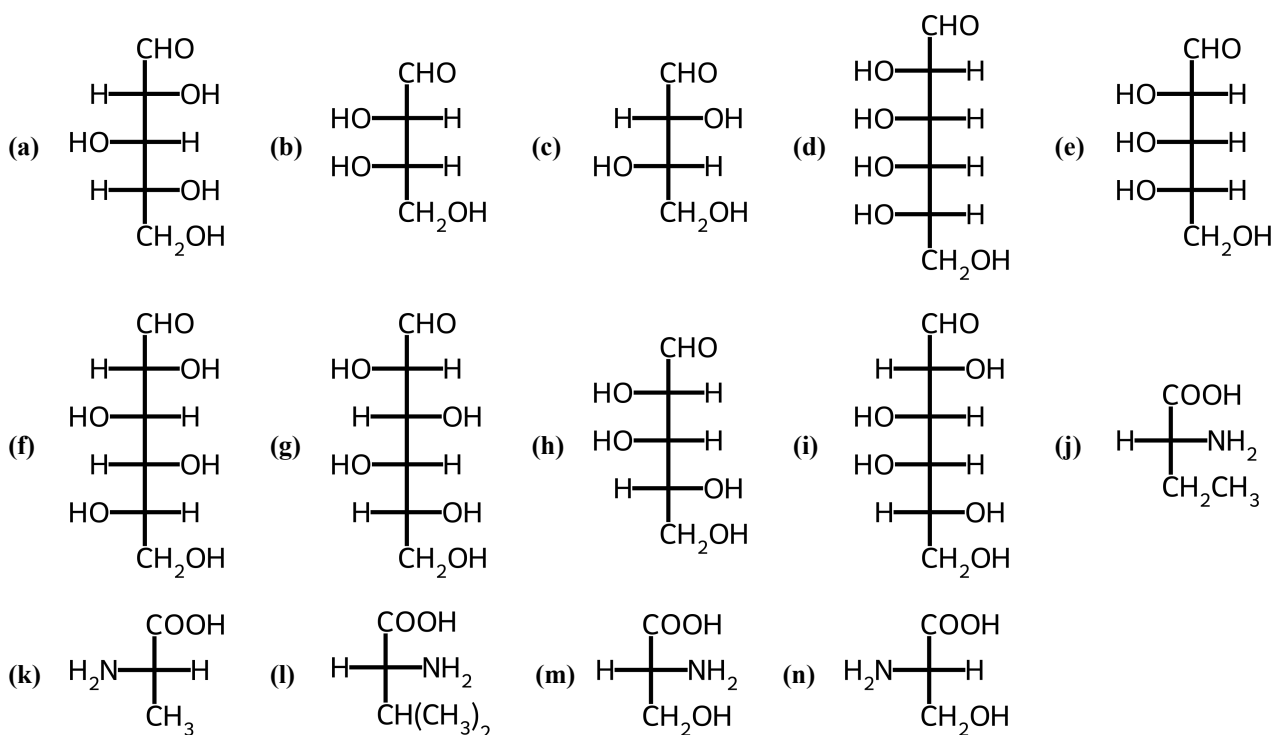
24. How many of the following compounds have 'R' configuration at the stereocentre (Cahn–Ingold–Prelog)?



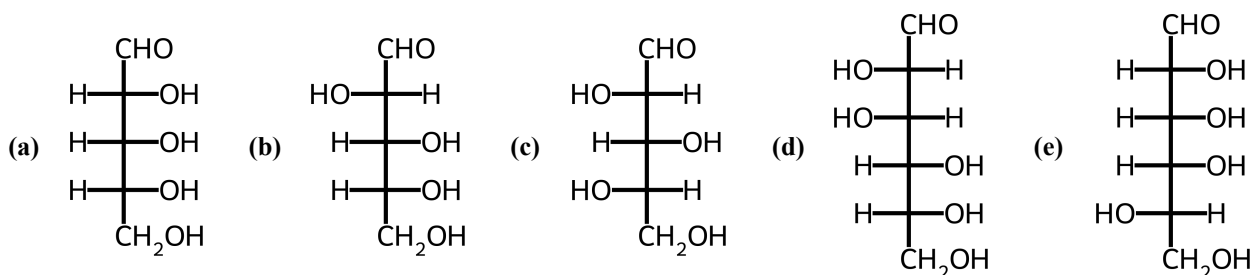
25. How many of the following compounds have 'S' configuration at the stereocentre (Cahn–Ingold–Prelog)?

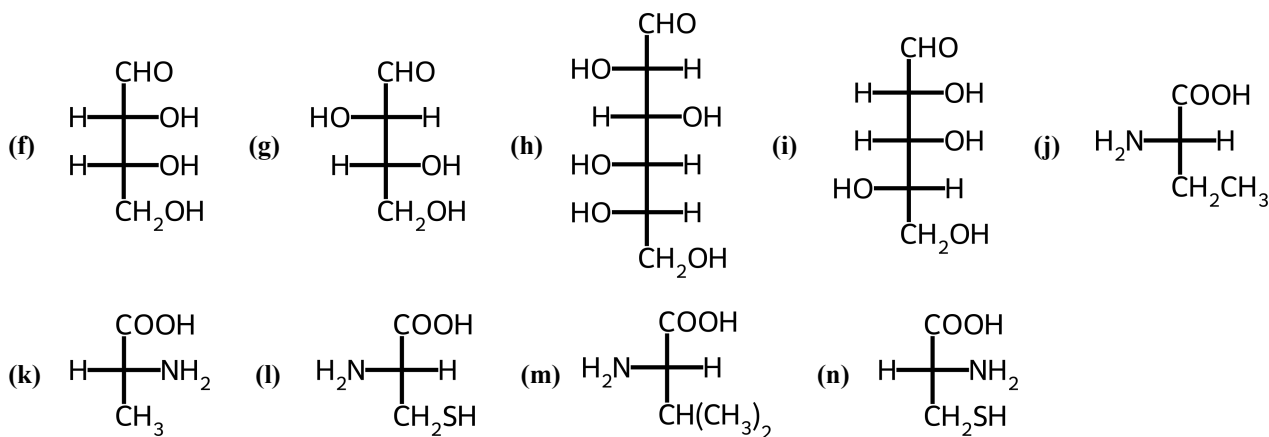


26. How many of the following Fischer projections represent D-configuration (reference: the stereocentre farthest from the top (most oxidised) carbon; $-\text{OH}/-\text{NH}_2$ on the right side)?

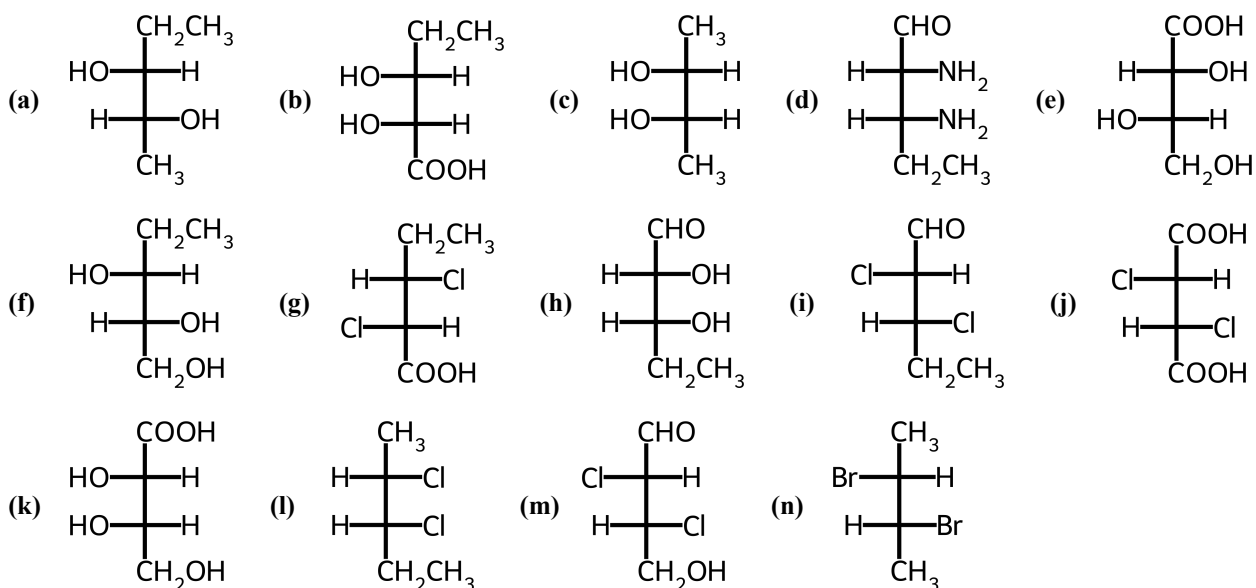


27. How many of the following Fischer projections represent L-configuration (reference: the stereocentre farthest from the top (most oxidised) carbon; $-\text{OH}/-\text{NH}_2$ on the left side)?

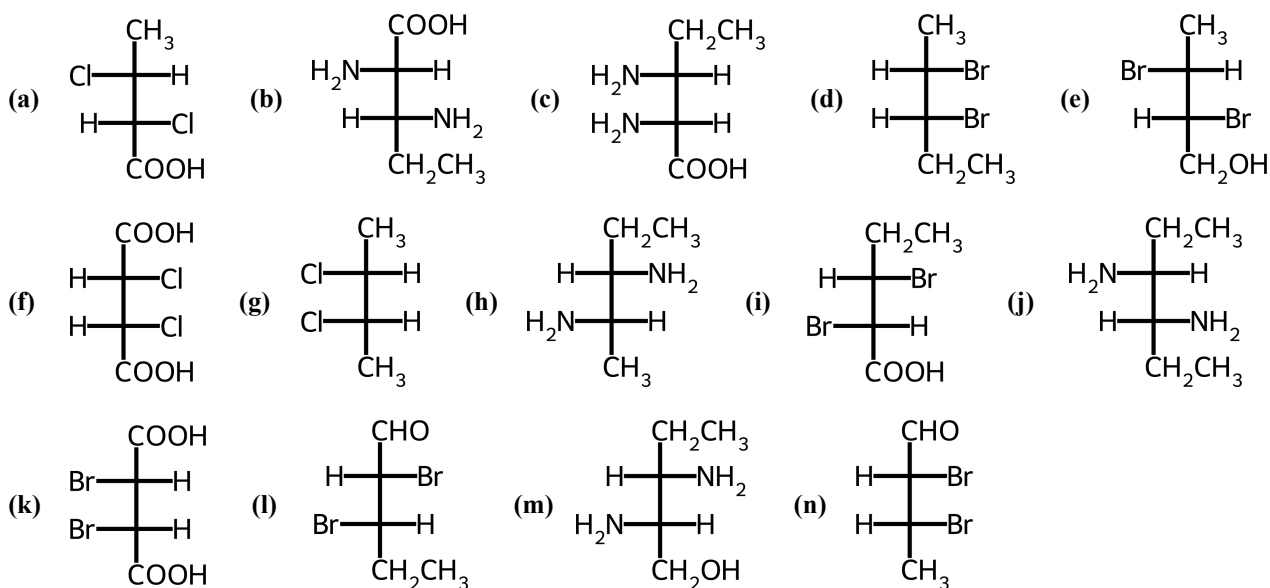




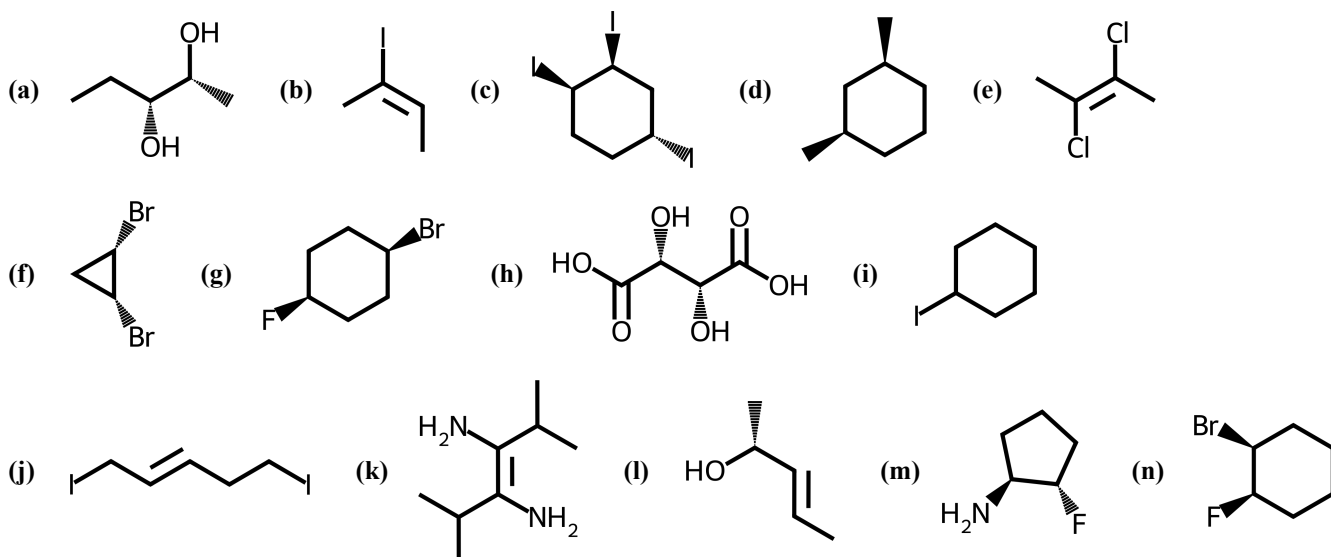
28. How many of the following Fischer projections are ERYTHRO isomers? (Erythro: the two similar (non-hydrogen) substituents on the two adjacent stereocentres lie on the SAME side of the vertical chain.)



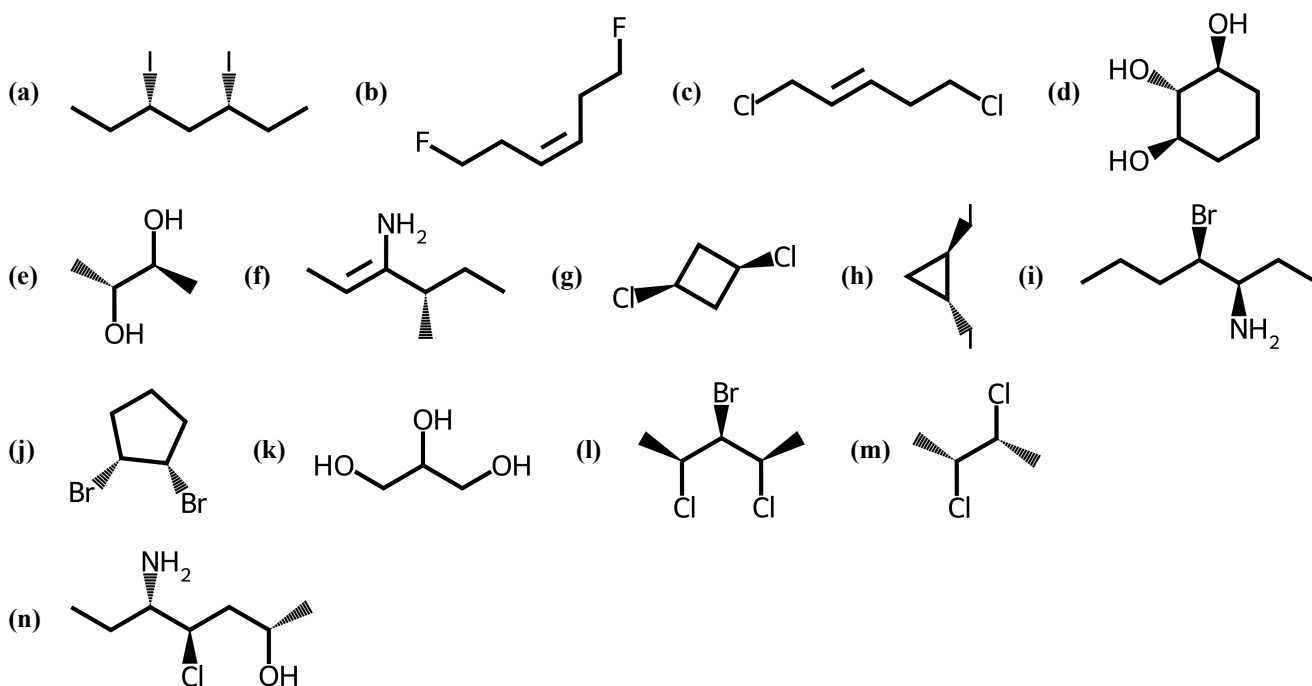
29. How many of the following Fischer projections are THREO isomers? (Threo: the two similar (non-hydrogen) substituents on the two adjacent stereocentres lie on OPPOSITE sides of the vertical chain.)



30. How many of the following compounds can be resolved into enantiomers at room temperature (stable, separable enantiomers exist for the drawn species)?

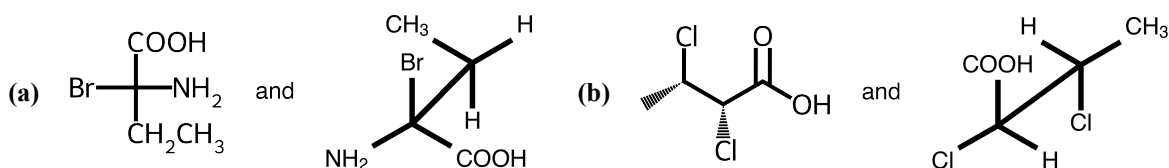


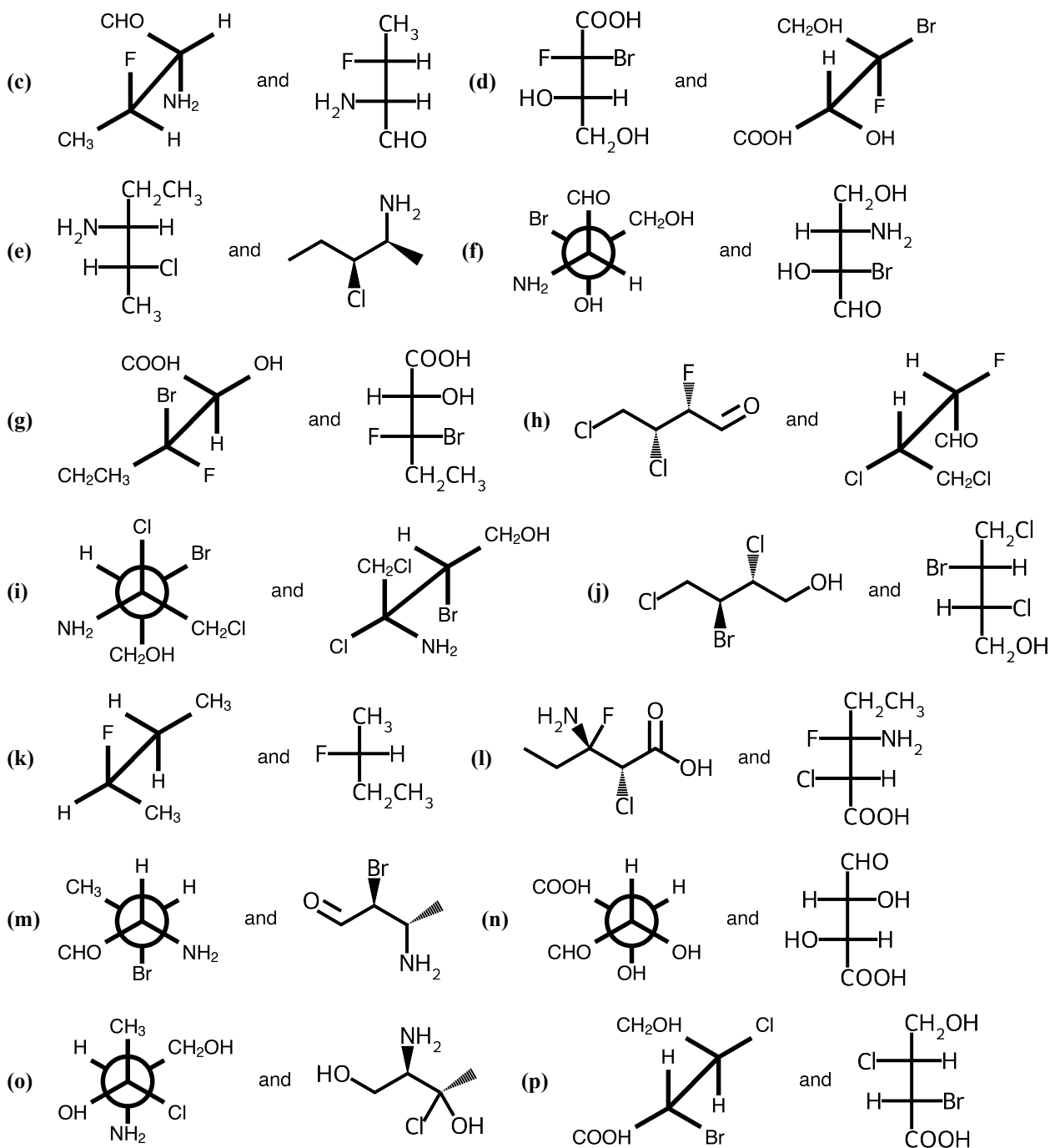
31. How many of the following molecules are meso compounds?



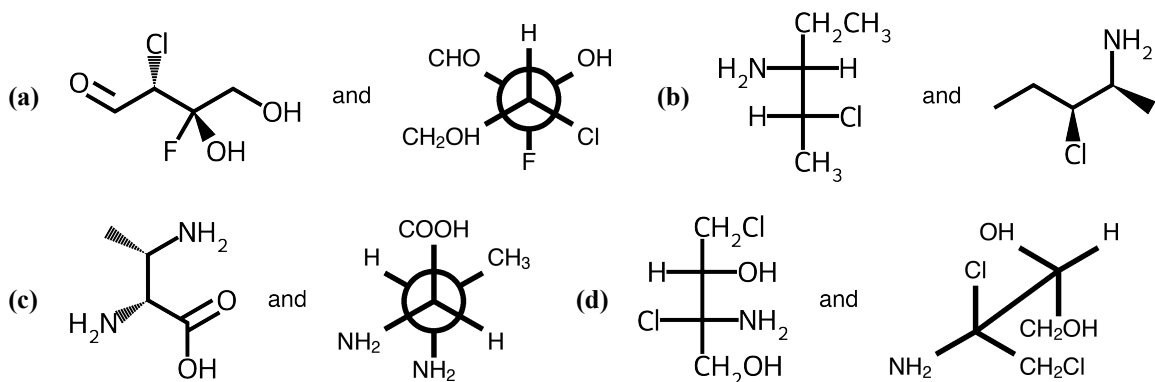
Part C — Relationship between pairs (all projection formulas)

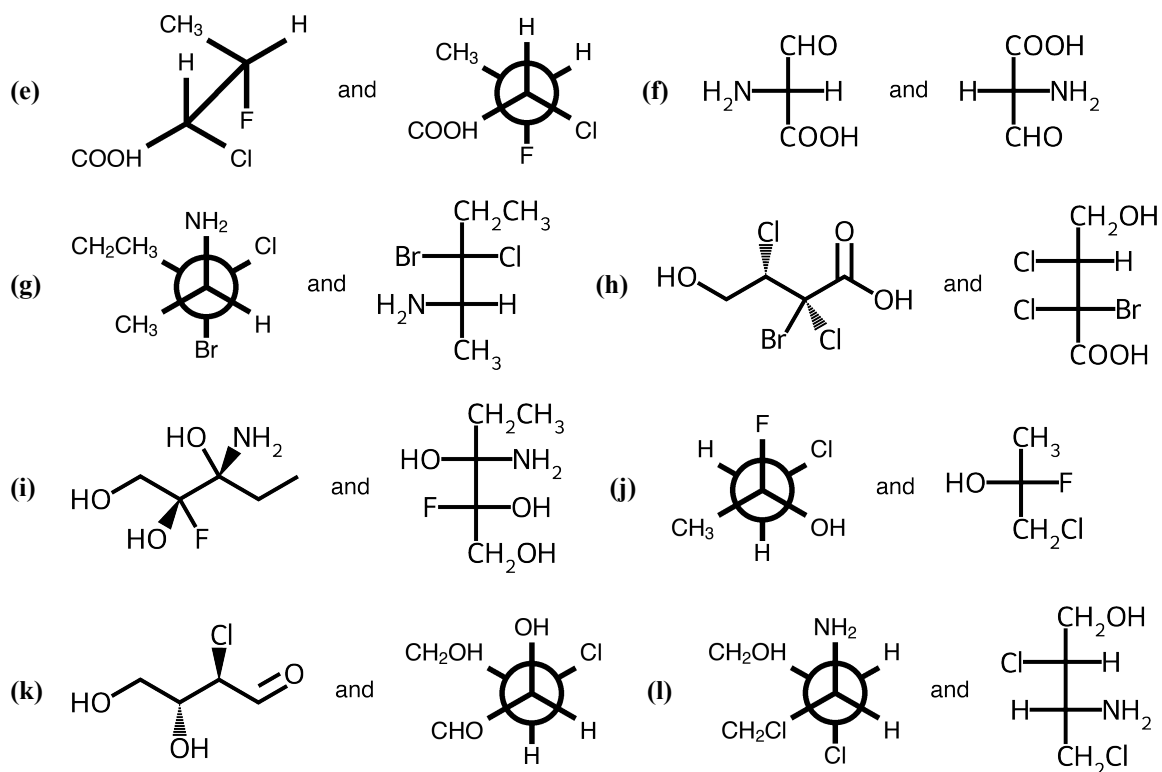
32. Each of the following pairs is drawn in two different projection formulas (Fischer, zig-zag wedge/dash, Newman, sawhorse). A Fischer projection may be turned only through 180° in the plane; Newman and sawhorse conformations may differ. Let a, b, c and d be the numbers of pairs that are ENANTIOMERS, DIASTEREOMERS, IDENTICAL molecules (homomers) and CONSTITUTIONAL isomers respectively. Find the four-digit number abcd.



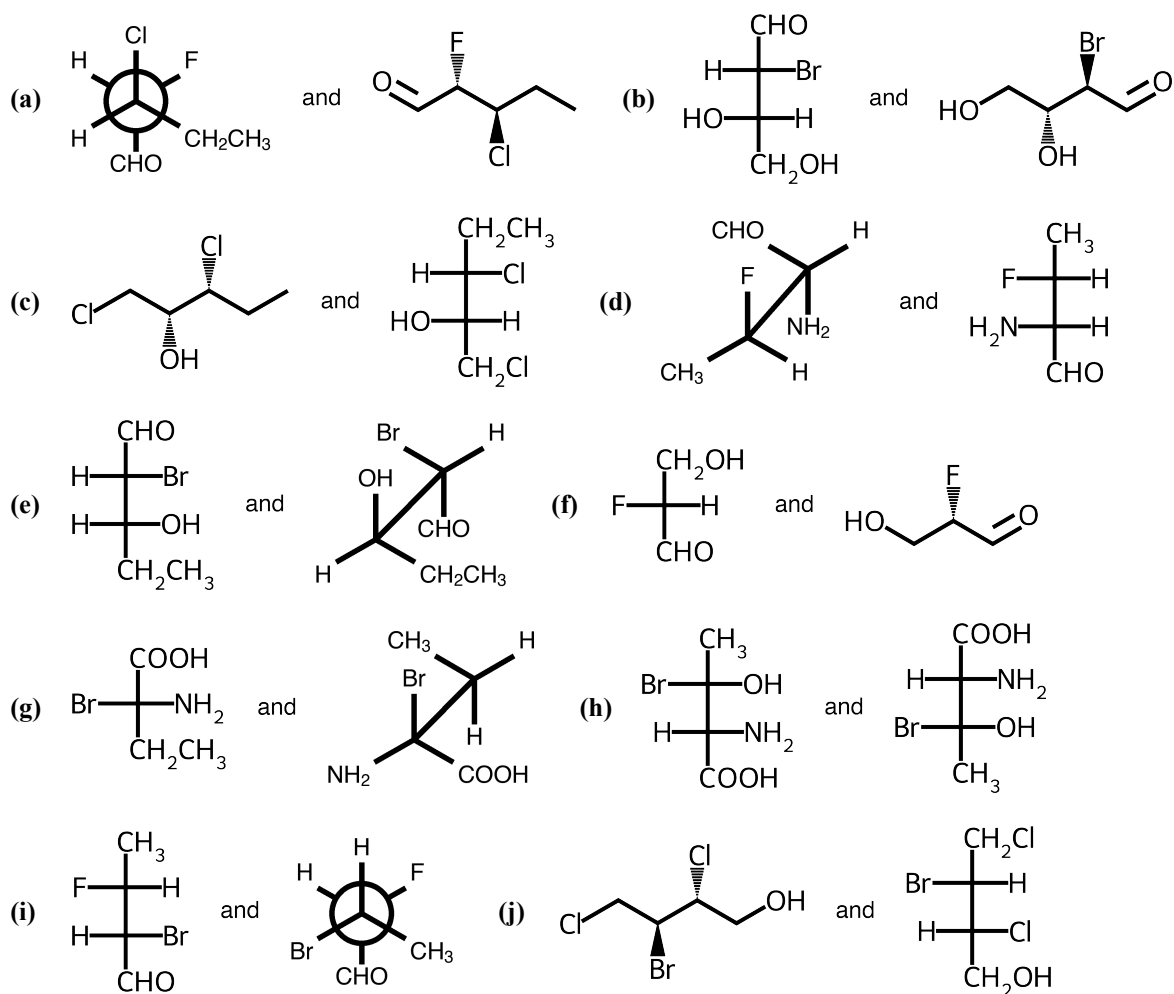


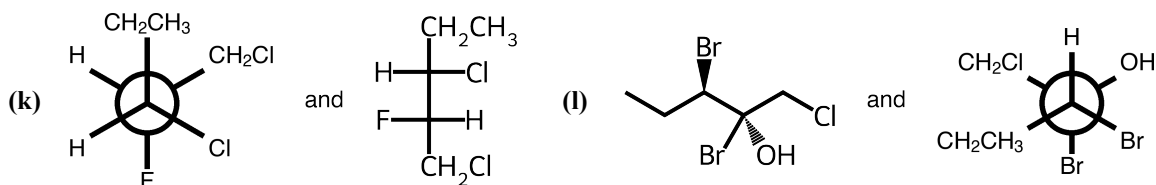
33. How many of the following pairs can, in principle, be separated by simple fractional distillation? (Assume that compounds with different physical constants in an achiral medium can be separated.)



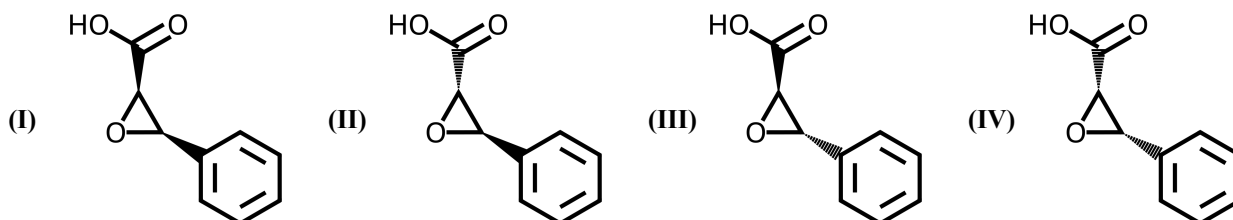


34. In how many of the following pairs do both members necessarily have the same boiling point, melting point, vapour pressure, density and solubility in an achiral solvent?

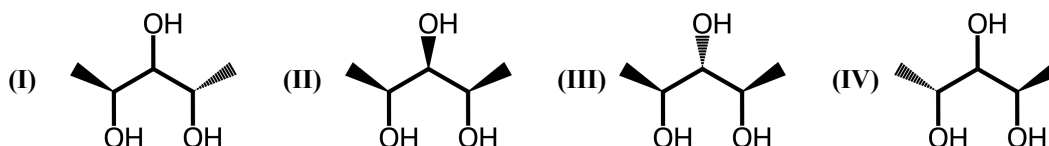




35. The four stereoisomers I–IV of the same compound are drawn below. Among the six possible pairs (each unordered pair counted once), how many are pairs of **DIASTEREOMERS**?

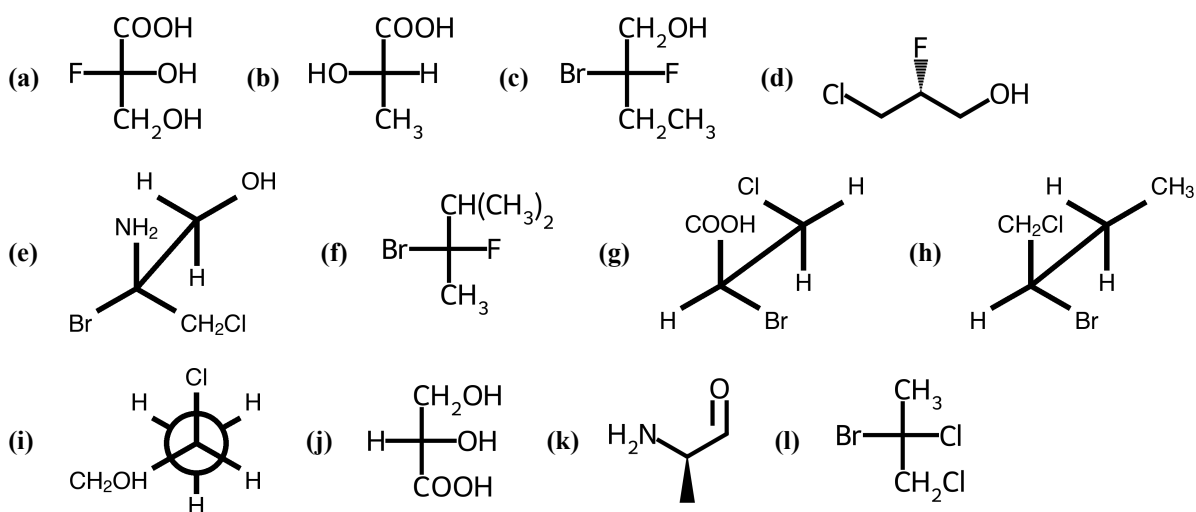


36. The four stereoisomers I–IV of the same compound are drawn below. Among the six possible pairs (each unordered pair counted once), how many are pairs of **ENANTIOMERS**?

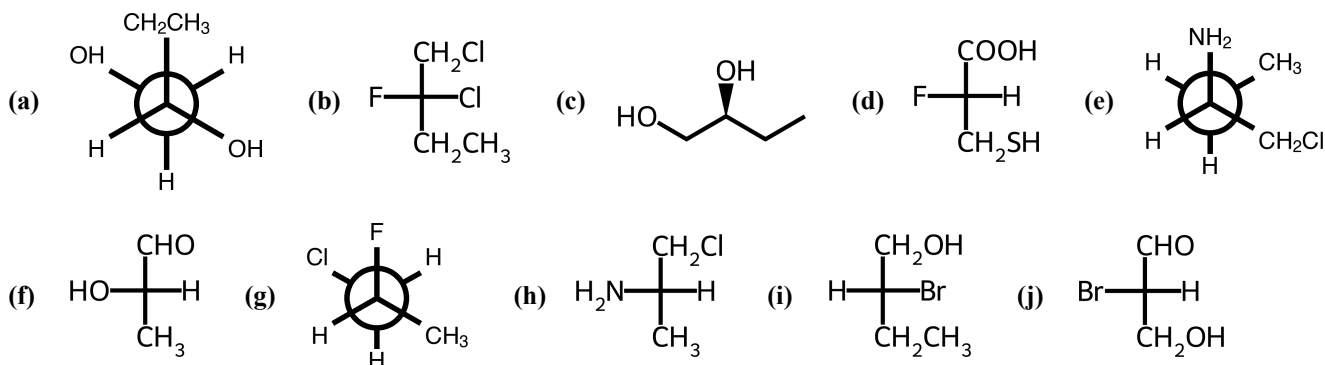


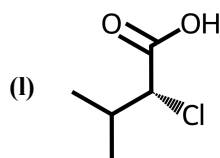
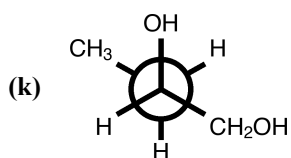
Part D — R/S, D/L and erythro/threo in different projections

37. How many of the following compounds, drawn in different projection formulas, have 'R' configuration at the stereogenic carbon? (Newman and sawhorse show the stereogenic carbon in front.)

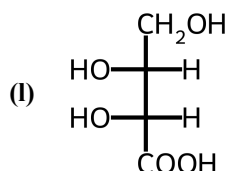
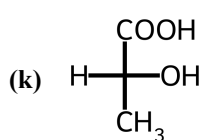
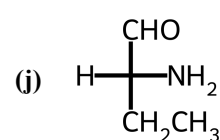
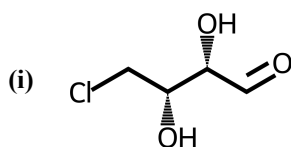
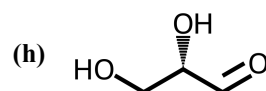
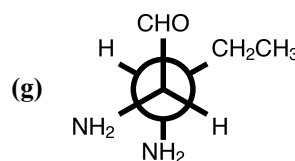
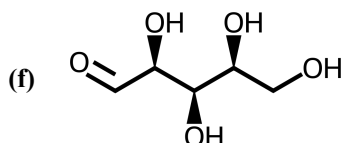
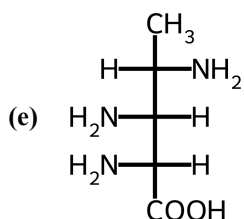
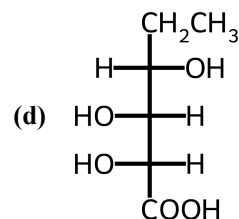
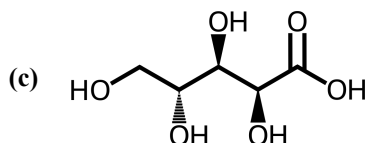
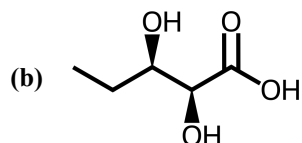
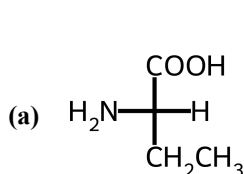


38. How many of the following compounds, drawn in different projection formulas, have 'S' configuration at the stereogenic carbon? (Newman and sawhorse show the stereogenic carbon in front.)

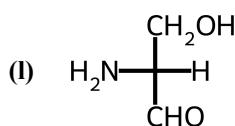
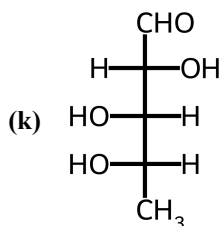
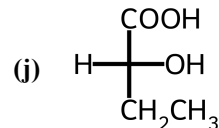
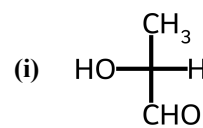
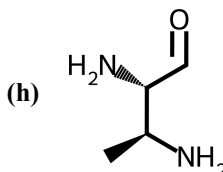
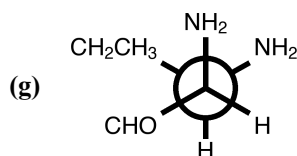
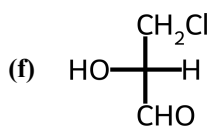
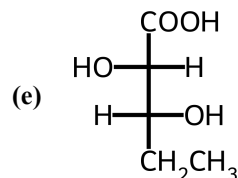
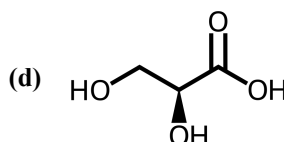
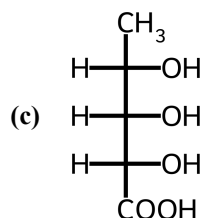
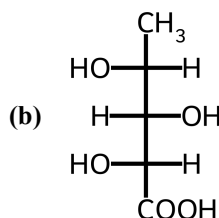
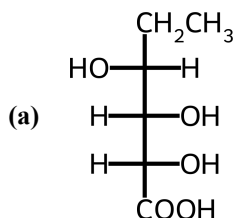




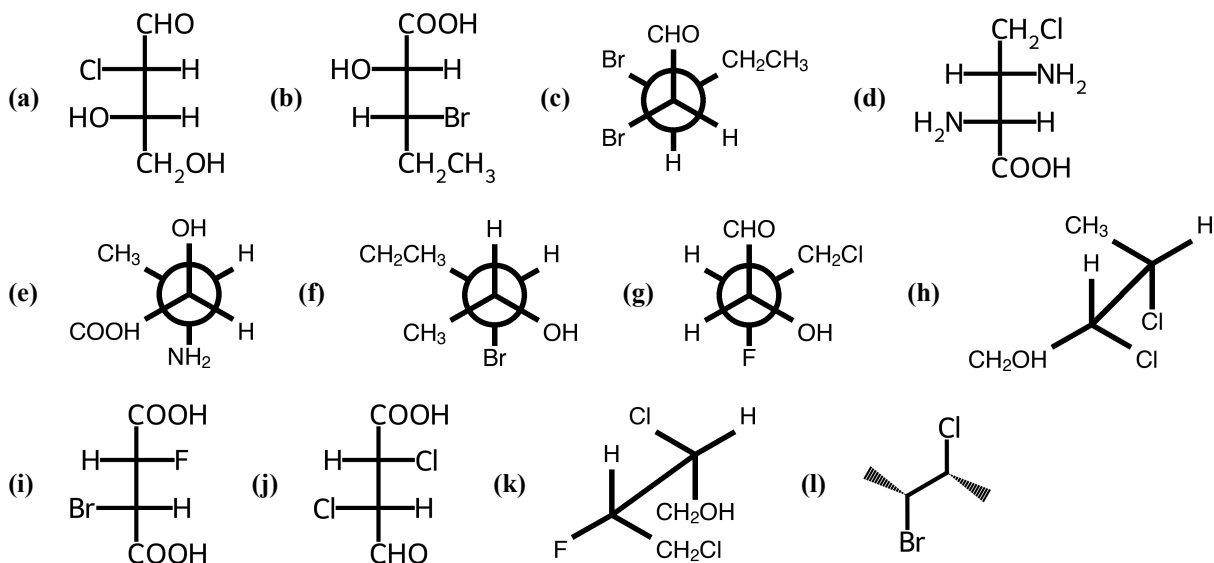
39. How many of the following compounds, drawn in different projection formulas, have D-configuration? (Convert mentally to a Fischer projection with the most oxidised carbon, $-\text{CHO}$ or $-\text{COOH}$, at the top; the D/L label is decided by the stereocentre farthest from it.)



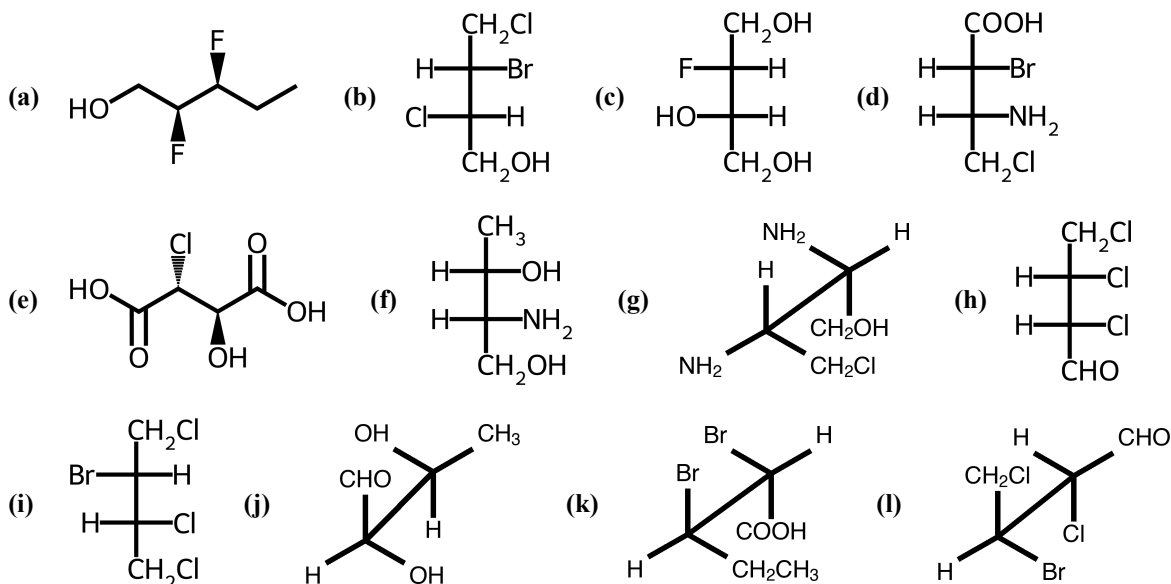
40. How many of the following compounds, drawn in different projection formulas, have L-configuration? (Convert mentally to a Fischer projection with the most oxidised carbon, $-\text{CHO}$ or $-\text{COOH}$, at the top; the L/L label is decided by the stereocentre farthest from it.)



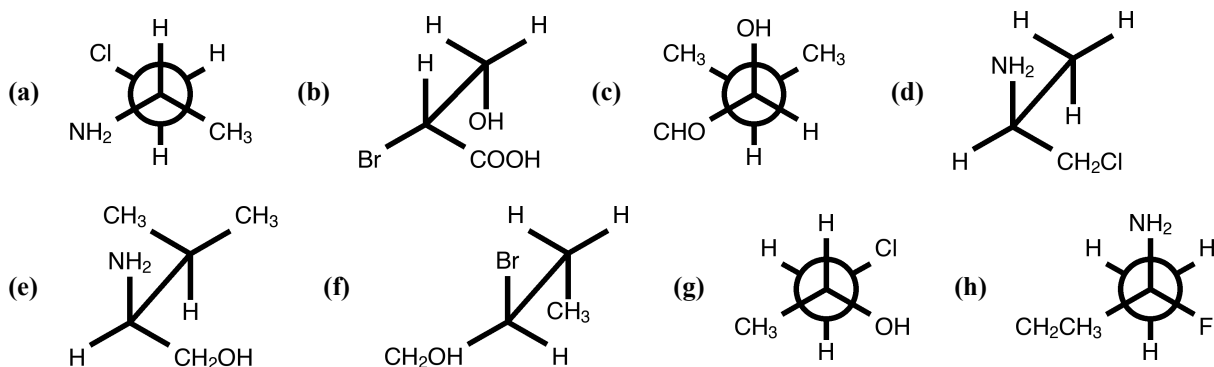
41. How many of the following compounds, drawn in different projection formulas, are ERYTHRO isomers? (Convert to a Fischer projection: the two non-hydrogen substituents on the adjacent stereocentres lie on the same side for erythro.)

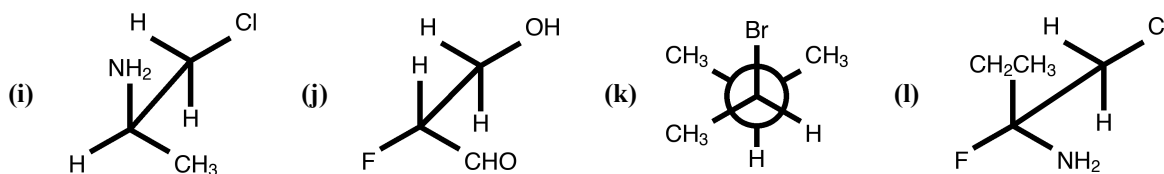


42. How many of the following compounds, drawn in different projection formulas, are THREO isomers? (Convert to a Fischer projection: the two non-hydrogen substituents on the adjacent stereocentres lie on the opposite side for threo.)

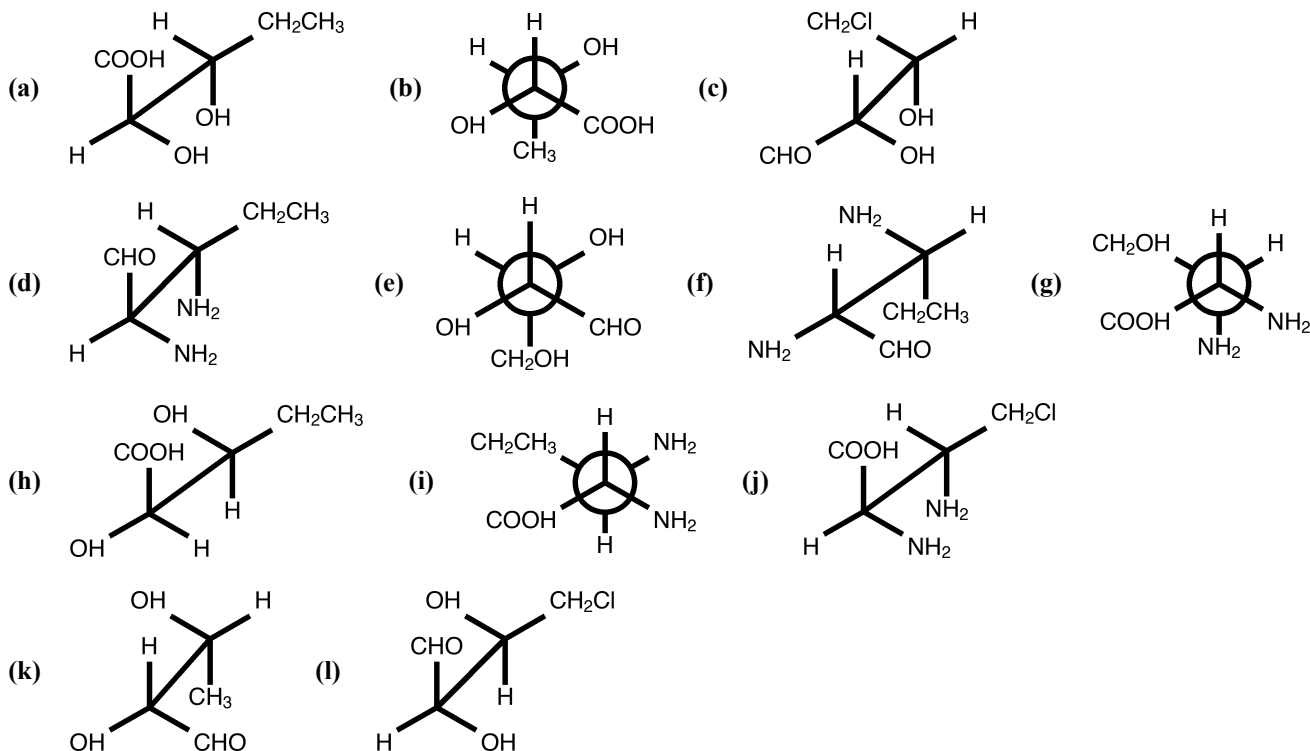


43. How many of the following compounds, drawn as Newman or sawhorse projections, have 'R' configuration at the stereogenic carbon? (Newman and sawhorse show the stereogenic carbon in front.)

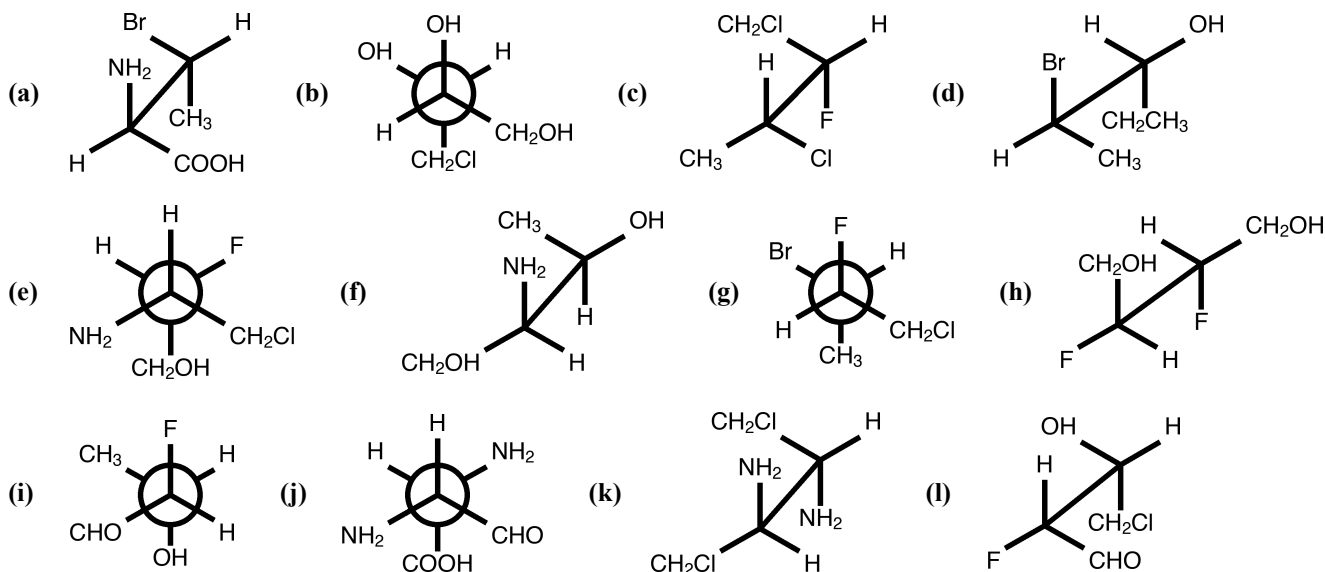




44. How many of the following compounds, drawn as Newman or sawhorse projections, have D-configuration? (Convert mentally to a Fischer projection with the most oxidised carbon, $-\text{CHO}$ or $-\text{COOH}$, at the top; the D/L label is decided by the stereocentre farthest from it.)

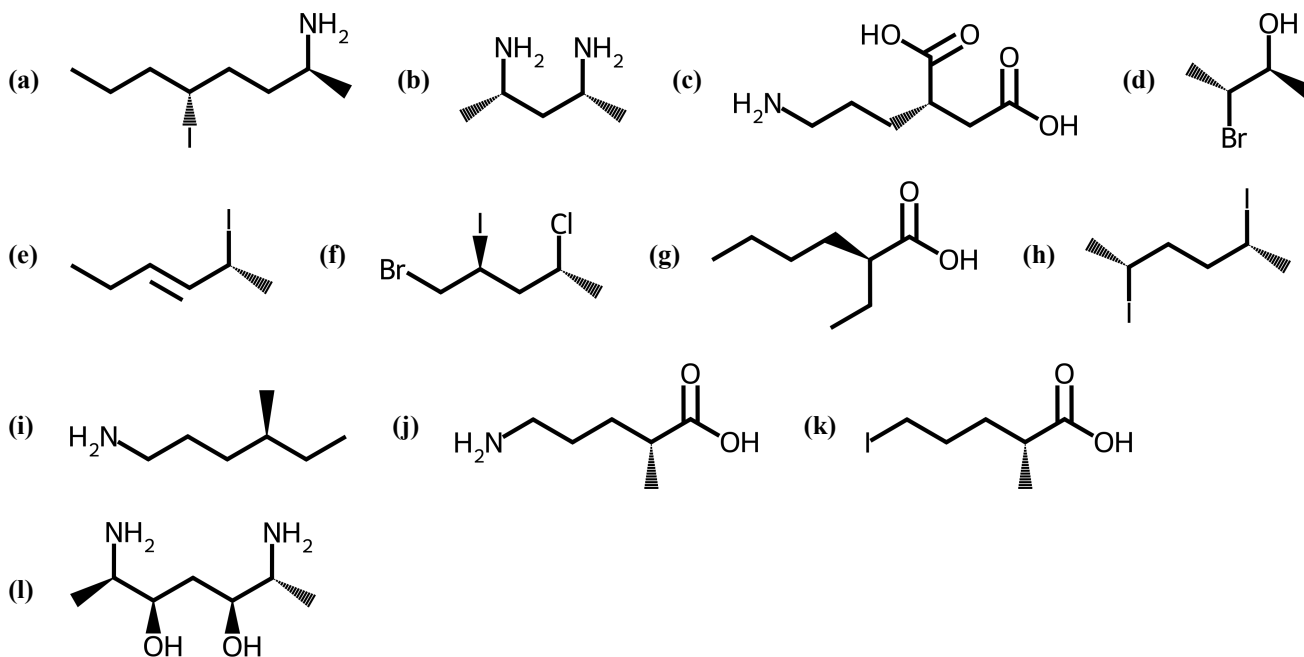


45. How many of the following compounds, drawn as Newman or sawhorse projections, are ERYTHRO isomers? (Convert to a Fischer projection: the two non-hydrogen substituents on the adjacent stereocentres lie on the same side for erythro.)

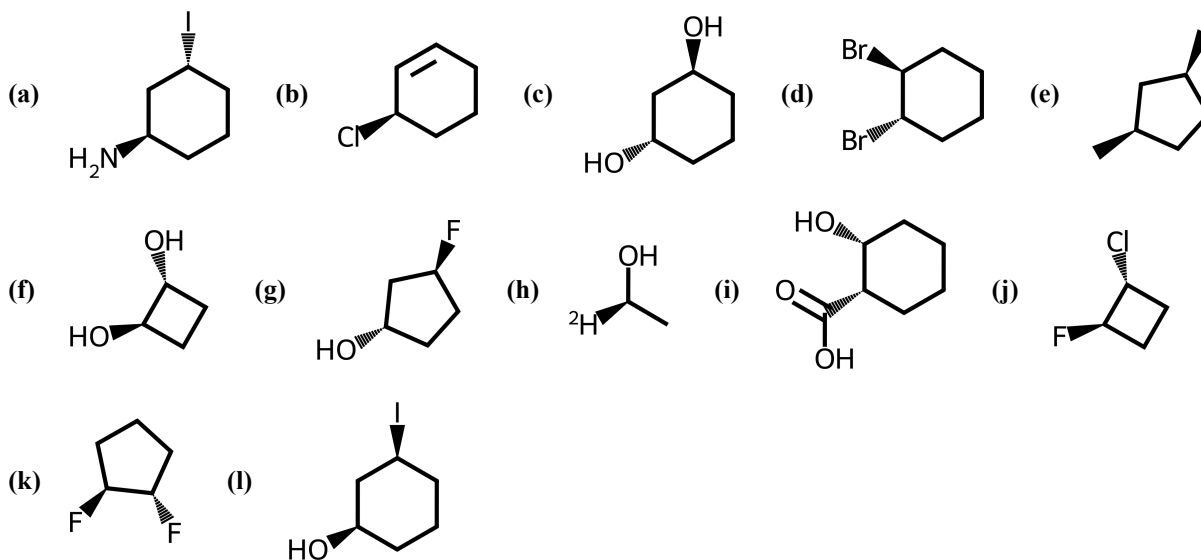


Part E — Chiral centres and substitution products

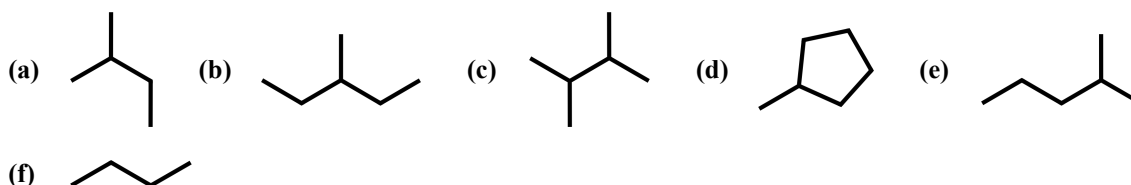
46. Find the total number of chiral (asymmetric) carbon centres present in the following acyclic compounds (a chiral centre = carbon bonded to four constitutionally different groups).



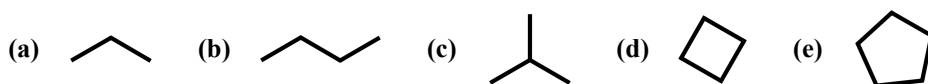
47. Find the total number of chiral (asymmetric) carbon centres present in all the following compounds (a chiral centre = carbon bonded to four constitutionally different groups).



48. Find the total number of OPTICALLY ACTIVE MONOCHLORO derivatives (structural isomers and stereoisomers, including both enantiomers) obtainable by replacing hydrogen atoms with chlorine, summed over: (a) 2-methylbutane; (b) 3-methylpentane; (c) 2,3-dimethylbutane; (d) methylcyclopentane; (e) 2-methylpentane; (f) butane.

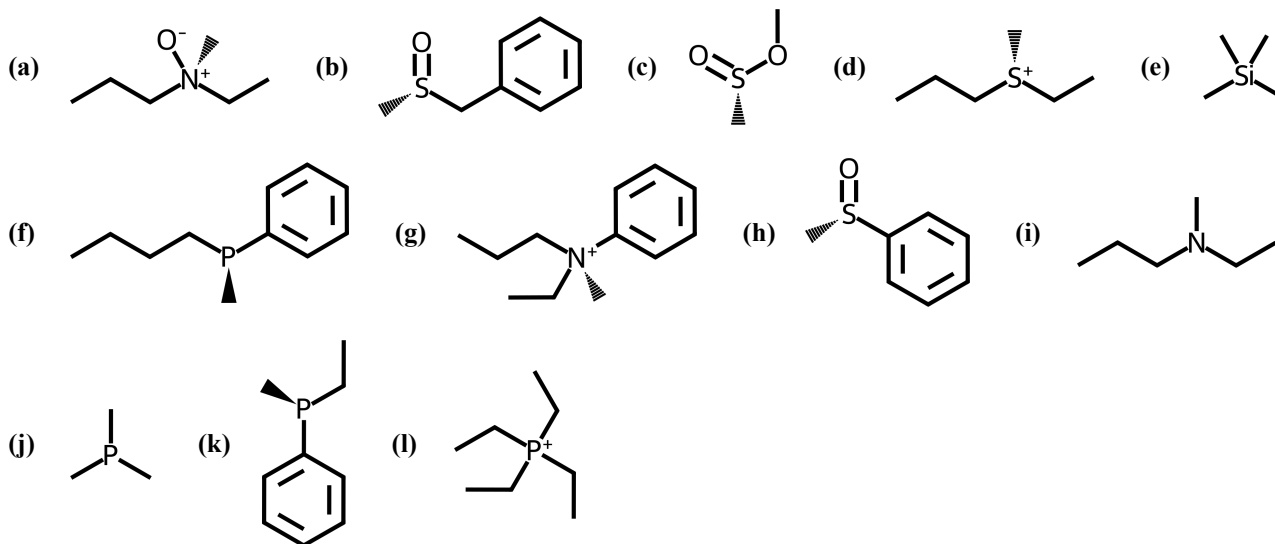


49. Find the total number of OPTICALLY ACTIVE DICHLORO derivatives (structural isomers and stereoisomers, including both enantiomers) obtainable by replacing hydrogen atoms with chlorine, summed over: (a) propane; (b) butane; (c) 2-methylpropane; (d) cyclobutane; (e) cyclopentane.

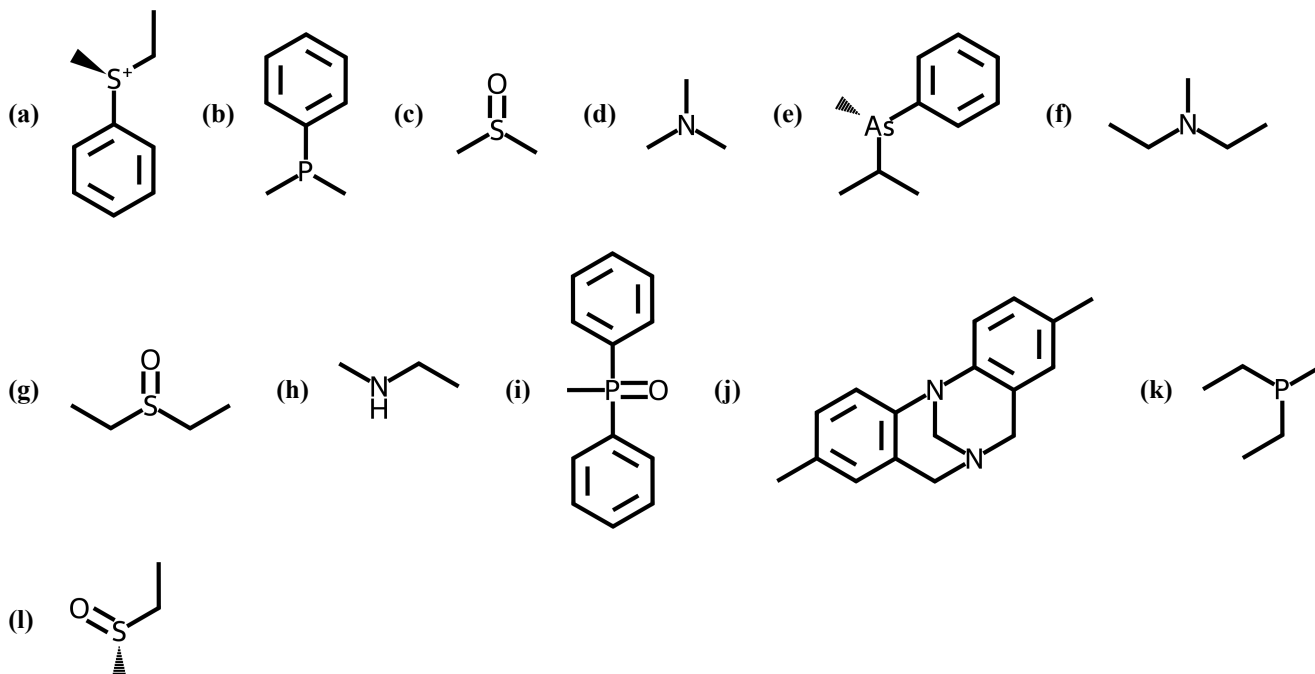


Part F — Stereocentres other than carbon

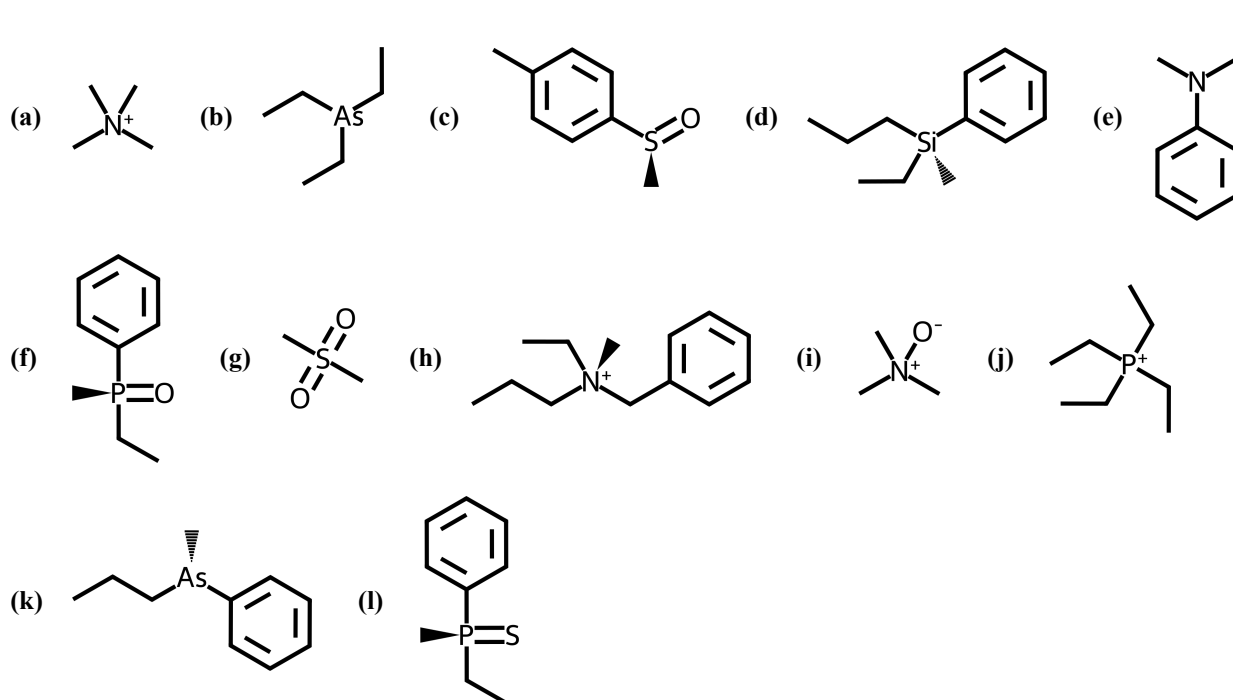
50. How many of the following species are chiral (optically active as drawn, each stereocentre being one enantiomer)? Treat nitrogen with three substituents + lone pair as freely inverting.



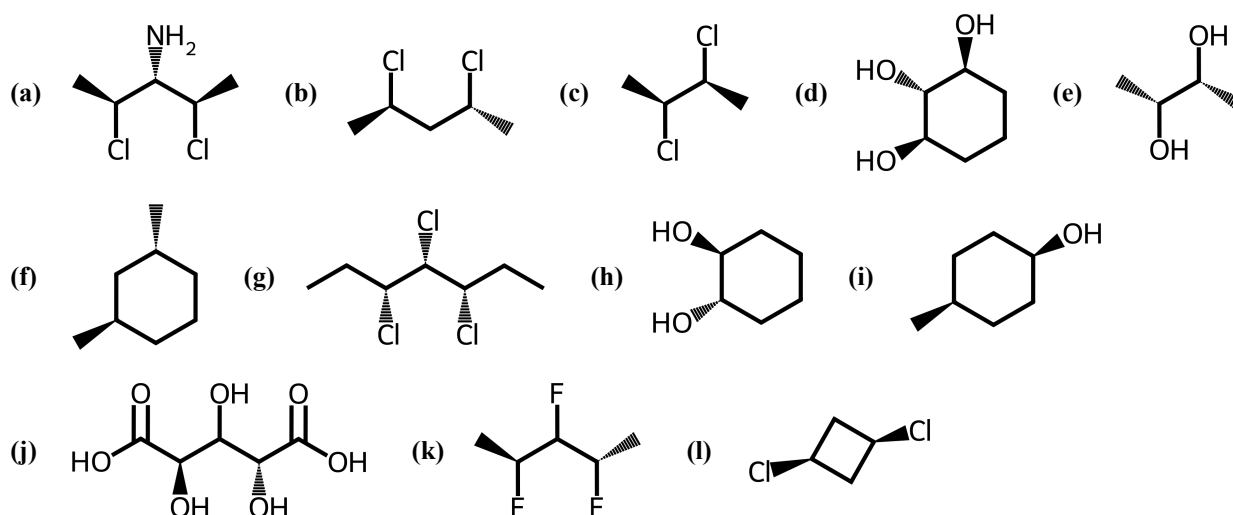
51. How many of the following can be resolved into stable enantiomers at room temperature?



52. How many of the following are chiral because of a stereogenic atom of group 15 (N, P or As) — not S or Si?

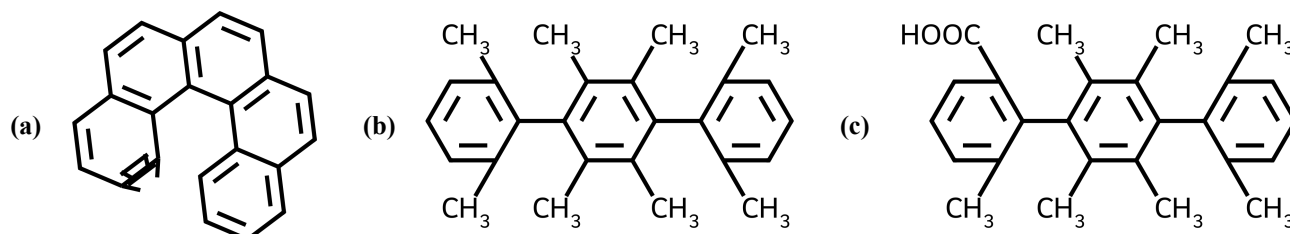


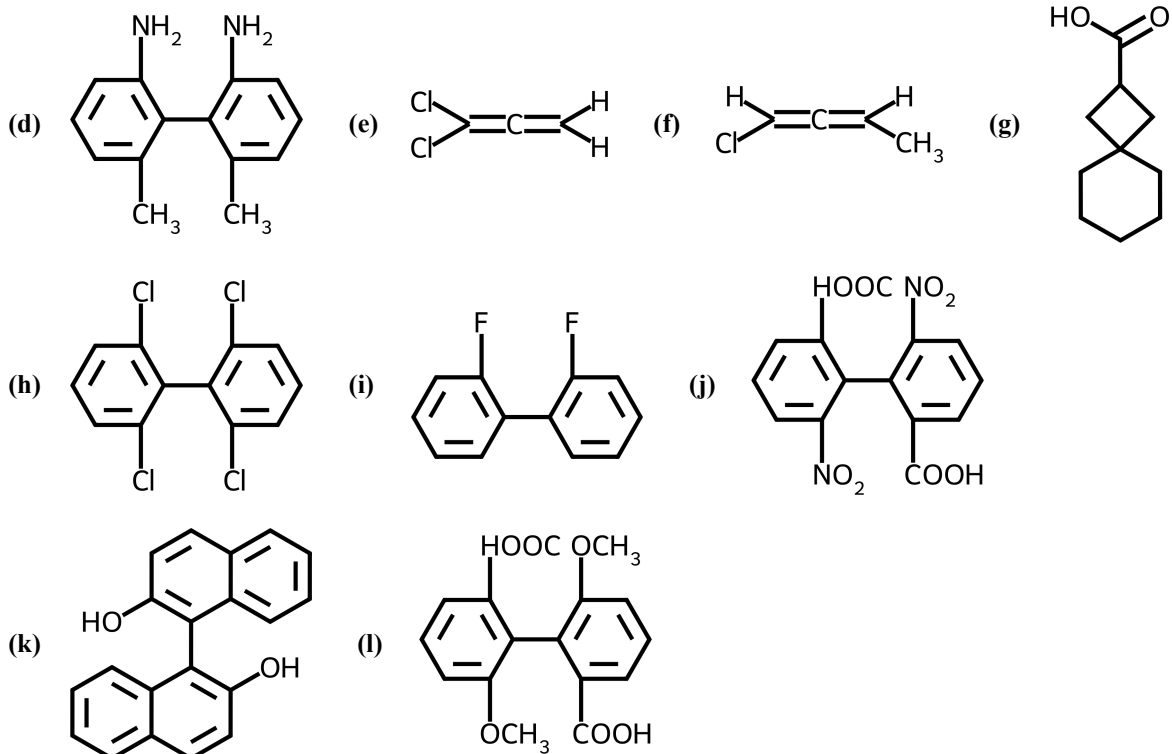
53. How many of the following structures contain a PSEUDOASYMMETRIC centre (a stereogenic atom bearing two constitutionally identical but oppositely configured groups, labelled r or s)?



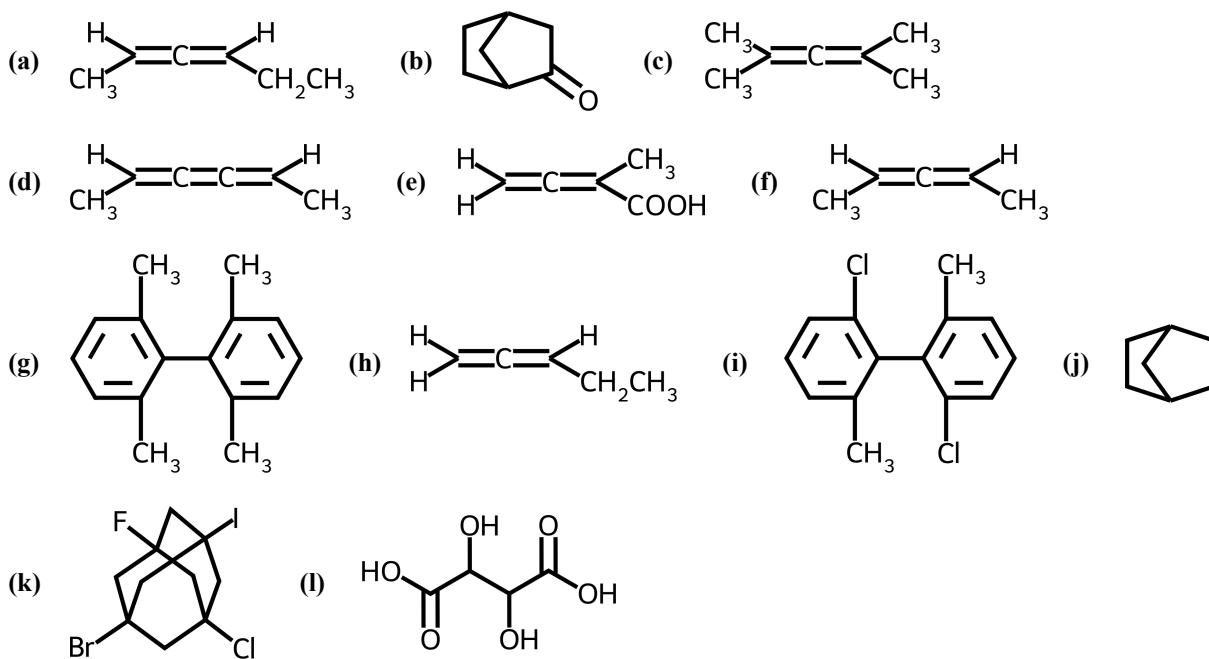
Part G — Spiranes, bicyclics, allenes, biphenyls, terphenyls

54. How many of the following can show optical isomerism (some stereoisomer of the skeleton drawn is optically active)?

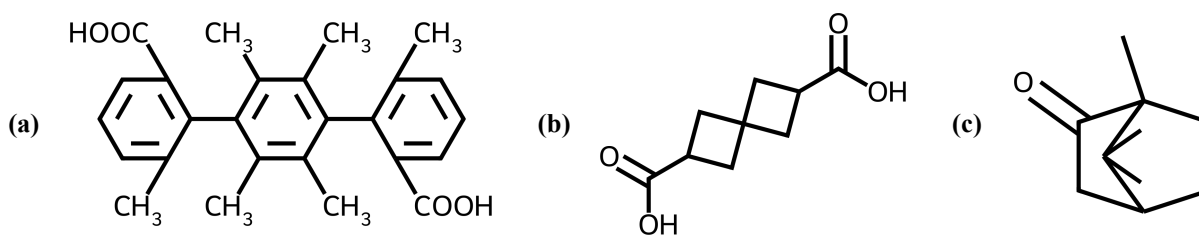


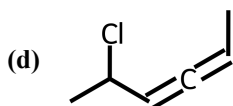


55. How many of the following can be optically active even though they contain NO stereogenic (asymmetric) carbon atom at all?

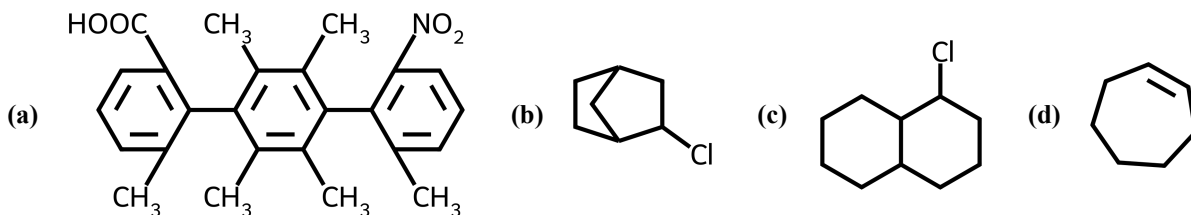


56. Find the total number of stereoisomers possible for all the compounds (a), (b), (c) and (d) taken together.

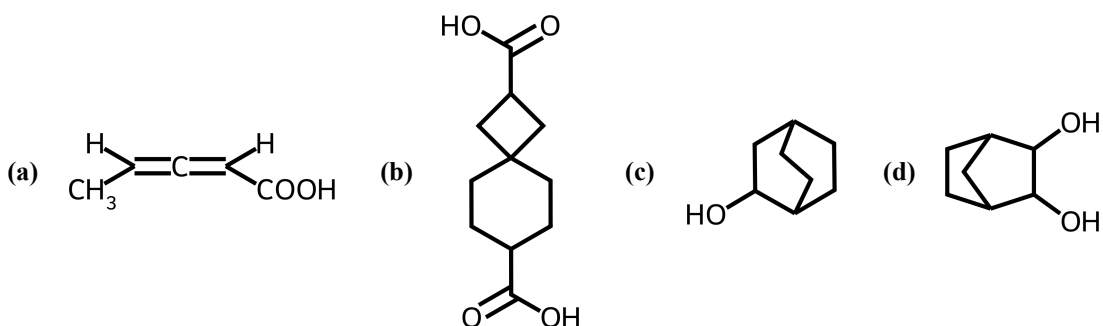




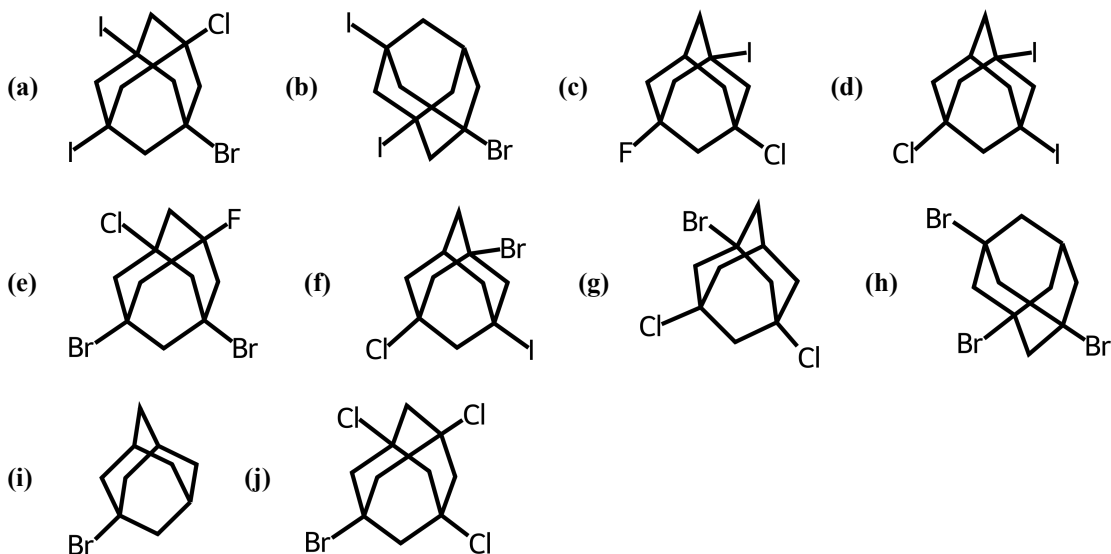
57. If a, b, c and d are the numbers of stereoisomers of compounds (a), (b), (c) and (d) respectively, find $a + b + c + d$.



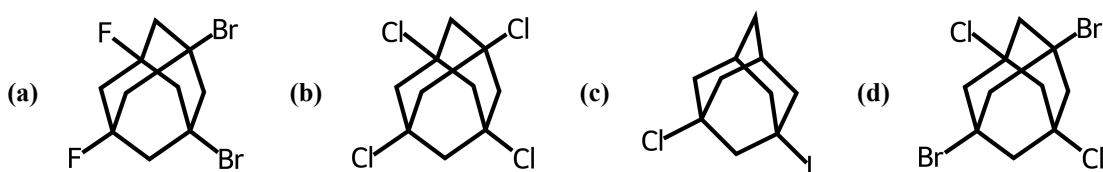
58. Find the total number of stereoisomers of (a), (b), (c) and (d) taken together.

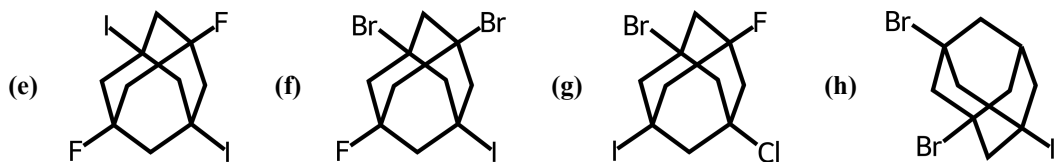


59. Each adamantane carries groups on its four bridgehead carbons (H where nothing is drawn). How many of the following are optically active?

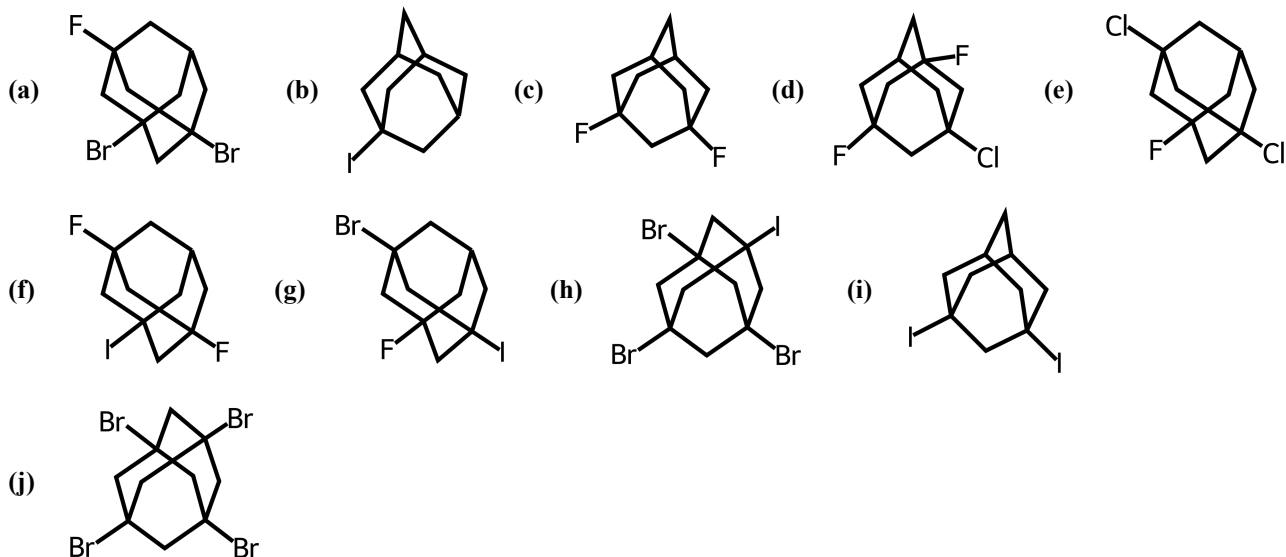


60. Each adamantane carries groups on its four bridgehead carbons (H where nothing is drawn). Mark POS, COS, AOS and AAOS for each compound and find the total number of marks over all eight compounds.



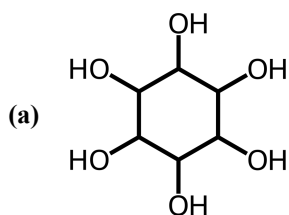


61. Each adamantane carries groups on its four bridgehead carbons (H where nothing is drawn). How many of the following possess a proper axis of symmetry (C_n , $n \geq 2$)?

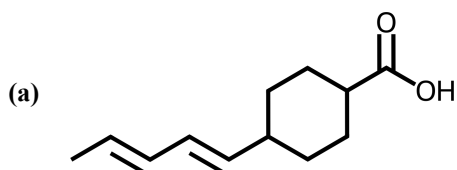


Part H — Counting stereoisomers

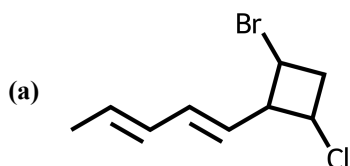
62. How many stereoisomers are possible for the following compound?



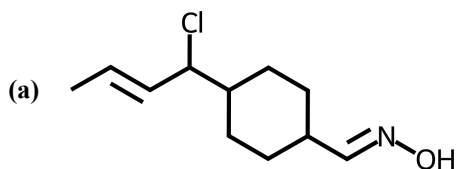
63. How many stereoisomers are possible for the following compound?



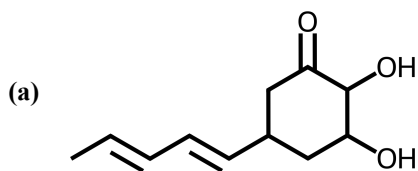
64. How many stereoisomers are possible for the following compound?



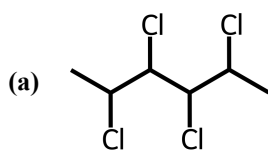
65. The total number of stereoisomers of the compound shown is 2^n . Find n.



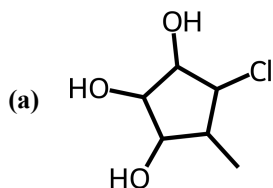
66. If the total number of stereoisomers of the compound shown is X, find $(X - 5)$.



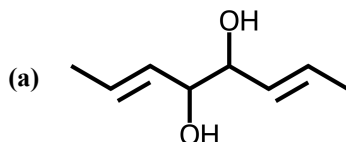
67. For the compound shown let x = total stereoisomers, y = optically active stereoisomers and z = optically inactive stereoisomers. Find $x + y + z$.



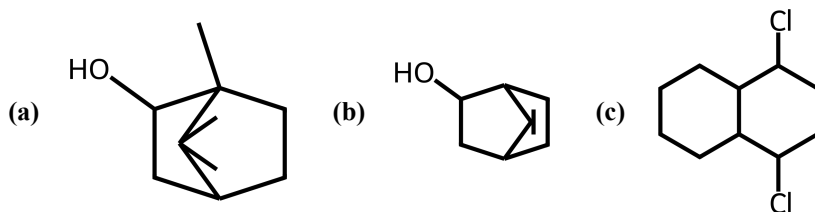
68. Find the number of pairs of enantiomers possible for the compound shown.



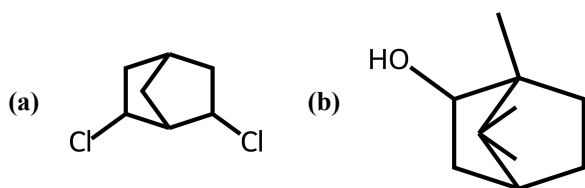
69. Find the number of optically INACTIVE stereoisomers of the compound shown.



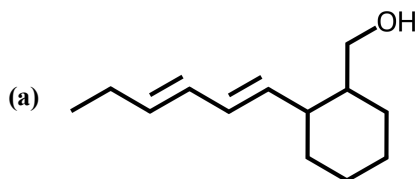
70. If a, b and c are the numbers of stereoisomers of the compounds (a), (b) and (c) shown, find $a + b + c$.



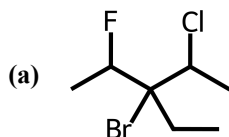
71. Find the total number of optically active stereoisomers of the compounds (a) 2,6-dichloronorbornane and (b) 7,7-dimethylnorbornan-2-ol taken together.



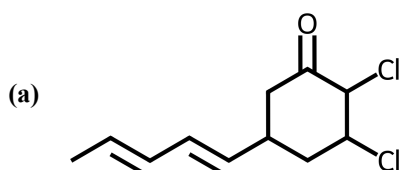
72. For the compound shown find (number of diastereomeric pairs) – (number of enantiomeric pairs) among all its stereoisomers (each unordered pair counted once).



73. Find the number of optically active stereoisomers of the compound shown.



74. Calculate $x + y + z$ where x = total stereoisomers, y = optically inactive stereoisomers and z = pairs of enantiomers, for the compound shown.



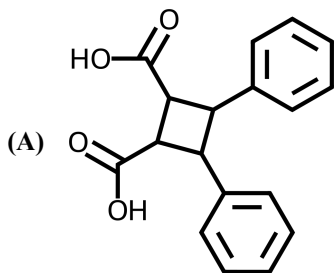
Part I — Optical rotation and optical purity

75. The specific rotation of sucrose is $+66.5^\circ$. What is the observed rotation (in degrees) of a solution containing 10 g of sucrose per 100 mL in a 20 cm polarimeter tube?
76. A solution of 5 g of an optically active compound in 100 mL of solvent, placed in a 10 cm tube, gives an observed rotation of $+2.4^\circ$. Find its specific rotation (in degrees).
77. The specific rotation of pure (+)-A is $+60^\circ$. Find the specific rotation of a mixture containing 80% (+)-A and 20% (–)-A.
78. 3 g of (+)-X and 1 g of (–)-X are dissolved together in a total volume of 20 mL and placed in a 20 cm tube. The specific rotation of pure (+)-X is $+40^\circ$. Find the observed rotation (in degrees).
79. A compound has specific rotation -52° . What is the observed rotation (in degrees) for a 0.4 g/mL solution in a 25 cm tube?
80. A partially racemised sample has $[\alpha] = +18^\circ$, while the pure (+)-enantiomer has $[\alpha] = +30^\circ$. What percentage of the sample is the (+)-enantiomer?
81. How many grams of (–)-enantiomer must be mixed with 6 g of the (+)-enantiomer to obtain a mixture of 50% optical purity?
82. α -D-glucose has $[\alpha] = +112^\circ$ and β -D-glucose has $[\alpha] = +19^\circ$. At mutarotation equilibrium the mixture contains 36% α and 64% β . Find the equilibrium specific rotation (in degrees).
83. A solution of 0.5 g/mL in a 10 cm tube rotates plane-polarised light by $+10^\circ$. What will be the rotation (in degrees) if the concentration is changed to 0.2 g/mL and a 25 cm tube is used?
84. 6 g of 2-butanol are dissolved in a total volume of 40 mL and filled in a 200 mm tube. The observed rotation is $+4.05^\circ$. Find the specific rotation of 2-butanol (in degrees).
85. A mixture of enantiomers shows $[\alpha] = +9^\circ$ while the pure enantiomer has $[\alpha] = +36^\circ$. Find its optical purity (%).
86. A sample has 40% optical purity and the pure enantiomer has $[\alpha] = +75^\circ$. What is the observed specific rotation (in degrees)?

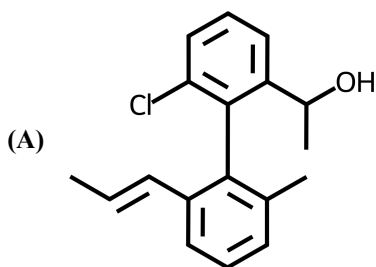
87. A sample of 2-butanol has $[\alpha] = -6.75^\circ$, whereas pure (–)-2-butanol has $[\alpha] = -13.5^\circ$. What percentage of the sample is racemic (equal amounts of both enantiomers)?
88. A mixture contains 9 mol of (R)- and 1 mol of (S)-enantiomer. Find the enantiomeric excess (%).

Part J — Stereoisomer calculations

89. Find the total number of open-chain stereoisomers of (A) an aldotetrose, (B) an aldopentose and (C) an aldohexose (all stereocentres unlike), i.e. $a + b + c$.
90. The compound shown (3,4-diphenylcyclobutane-1,2-dicarboxylic acid) exists in two or more stereoisomeric forms. Let a = total number of stereoisomers, b = number of pairs of enantiomers and c = number of meso compounds. Find $a + b + c$.

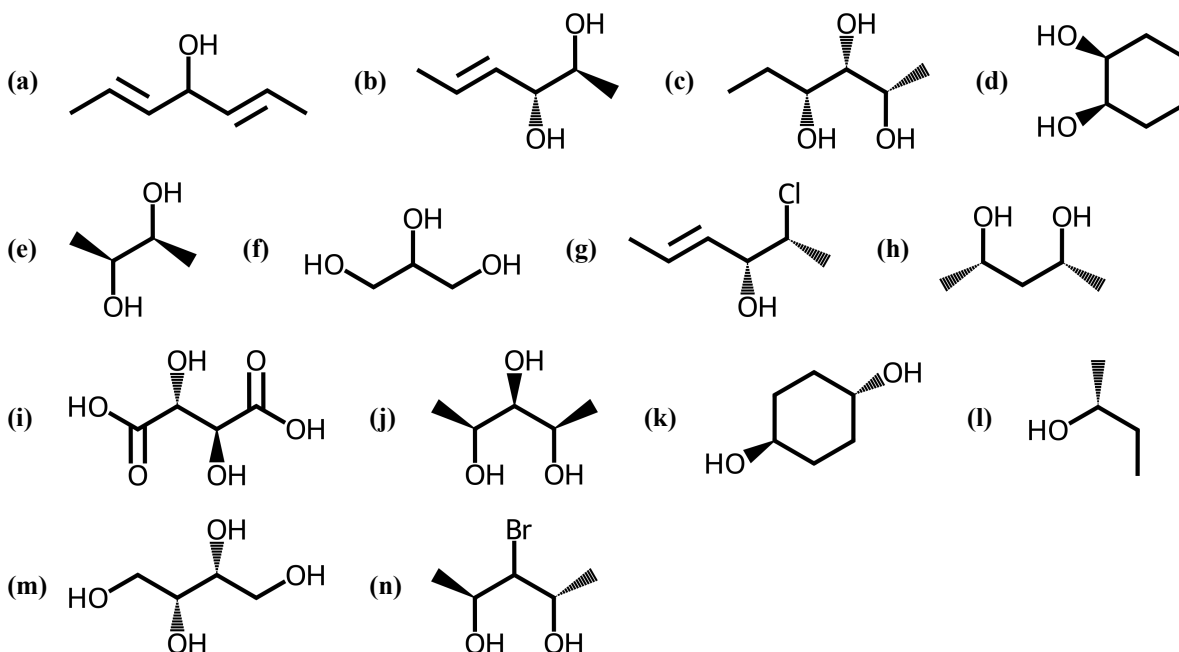


91. Find the total number of stereoisomers of the biaryl shown (hindered rotation about the aryl–aryl bond, one stereogenic carbon and one double bond).



Part K — Resolution and mixtures

92. Each alcohol drawn below is a single pure stereoisomer. How many of them give a mixture of DIASTEREOMERIC esters when treated with (±)-2-methylbutanoic acid?



93. Separate equimolar mixtures containing ALL the stereoisomers of (a) 2,3,4,5-tetrachlorohexane, (b) pentane-2,3,4-triol, (c) 2,3-dibromopentane and (d) 3,4-dichlorohexane are each fractionally distilled. Find the total number of fractions obtained for the four mixtures.
94. Each fraction of the mixtures in the previous question is passed through a chiral column. How many DIFFERENT optically active compounds are finally isolated from the four mixtures together?
95. Read the following 12 descriptions of mixtures and find in how many of them the mixture is optically INACTIVE.
- (1) an equimolar mixture of (+)- and (–)-lactic acid
 - (2) an equimolar mixture of (2R,3R)- and (2S,3S)-tartaric acid
 - (3) (+)-tartaric acid and meso-tartaric acid mixed in equal amounts
 - (4) 3 mol of (R)-butan-2-ol and 1 mol of (S)-butan-2-ol
 - (5) pure meso-tartaric acid
 - (6) an equimolar mixture of all four stereoisomers of 2,3-dichloropentane
 - (7) equimolar (R)-2-chlorobutane and (S)-2-bromobutane
 - (8) a racemic modification mixed with some pure (+)-enantiomer
 - (9) cis-1,2-dimethylcyclohexane, (+)-trans- and (–)-trans-1,2-dimethylcyclohexane in 1 : 1 : 1 ratio
 - (10) (R,R)-, (S,S)- and meso-2,3-dibromobutane in the mole ratio 1 : 2 : 3
 - (11) α -D-glucose and β -D-glucose mixed in equal amounts
 - (12) equal masses of (R)- and (S)-limonene
96. (2R,3R)-A has $[\alpha] = +30^\circ$, its enantiomer (2S,3S)-A has -30° and the meso diastereomer has 0° . A mixture contains 3 mol (R,R), 1 mol (S,S) and 2 mol meso isomer (all of equal molar mass). Find the specific rotation of the mixture (in degrees).
97. A compound has two optically active enantiomeric forms with $[\alpha] = \pm 45^\circ$ and a meso form. A sample is prepared by mixing 2 mol of (+), 3 mol of (–) and 5 mol of the meso form. Find the specific rotation (in degrees) of the sample (molar masses are equal).
98. Three-component mixture in a 10 cm tube: (+)-A ($[\alpha] = +60^\circ$) at 0.5 g/mL and (–)-A at 0.2 g/mL (same compound, so the (–)-form has $[\alpha] = -60^\circ$). Find the observed rotation (in degrees).

Part L — Statements on optical isomerism

99. Read the following 14 statements on optical activity and chirality and find how many of them are correct.
- (1) Every chiral molecule is, in principle, optically active.
 - (2) A tetrahedral carbon bonded to four different groups makes the molecule chiral when it is the only stereocentre.
 - (3) A molecule with two stereocentres is always chiral.
 - (4) The presence of a plane of symmetry makes a molecule achiral.
 - (5) A molecule having a centre of symmetry is achiral.
 - (6) A molecule that possesses a C_2 axis of symmetry cannot be chiral.
 - (7) The absence of every alternating (improper, S_n) axis of symmetry is the necessary and sufficient condition for chirality.
 - (8) A compound lacking a chiral carbon cannot be optically active.
 - (9) All optically active compounds contain at least one chiral carbon atom.
 - (10) A meso compound contains stereocentres but is optically inactive by internal compensation.
 - (11) The magnitude of the specific rotation of an enantiomer does not depend on the solvent or the temperature.
 - (12) Two enantiomers rotate plane-polarised light through equal angles but in opposite directions under identical conditions.
 - (13) A racemic mixture is optically inactive because of external compensation.
 - (14) The (+)/(–) sign of rotation directly tells whether a stereocentre is R or S.
100. Read the following 14 statements on symmetry, meso compounds and ring stereochemistry and find how many of them are correct.
- (1) meso-Tartaric acid has an internal plane of symmetry in its eclipsed conformation.
 - (2) All meso compounds are achiral.
 - (3) cis-1,2-Dimethylcyclohexane (taking the ring as planar on average) is a meso compound.

- (4) trans-1,2-Dimethylcyclohexane is optically active and exists as a pair of enantiomers.
 - (5) cis-1,3-Dimethylcyclohexane is a meso compound.
 - (6) trans-1,4-Dimethylcyclohexane is chiral.
 - (7) 2,3-Dibromobutane has three stereoisomers.
 - (8) (2R,3S)-2,3-Dibromobutane is chiral.
 - (9) Tartaric acid has four stereoisomers.
 - (10) A molecule having n dissimilar stereocentres has 2^n stereoisomers.
 - (11) An open-chain aldohexose such as glucose has 16 stereoisomers.
 - (12) The anti-staggered conformation of meso-2,3-dibromobutane has a centre of symmetry.
 - (13) Both stereocentres of meso-2,3-dibromobutane have the same CIP descriptor.
 - (14) A compound with a plane of symmetry can still contain chiral (stereogenic) centres.
101. Read the following 14 statements on CIP rules, Fischer projections and D/L and find how many of them are correct.
- (1) CIP priority is decided first by the atomic number of the atom directly attached to the stereocentre.
 - (2) $-\text{CH}_2\text{OH}$ has higher CIP priority than $-\text{COOH}$.
 - (3) The vinyl group ($-\text{CH}=\text{CH}_2$) has higher CIP priority than the isopropyl group.
 - (4) Interchanging any two groups on a stereocentre changes its descriptor from R to S or vice versa.
 - (5) An (S)-configured compound always rotates plane-polarised light in the $(-)$ direction.
 - (6) Rotating a Fischer projection by 90° in the plane of the paper inverts the configuration.
 - (7) Rotating a Fischer projection by 180° in the plane of the paper does not change the configuration.
 - (8) Every (R)-compound has D-configuration.
 - (9) In D-glyceraldehyde the $-\text{OH}$ group lies on the right in the Fischer projection and the centre is (R).
 - (10) Natural L-alanine is the (S)-enantiomer.
 - (11) In an erythro isomer the similar groups on adjacent stereocentres lie on opposite sides in the Fischer projection.
 - (12) D-(+)-Glucose has four chiral carbons, three of which are (R).
 - (13) Deuterium ($-\text{D}$) has higher CIP priority than hydrogen ($-\text{H}$).
 - (14) In CIP priority $-\text{SH}$ ranks above $-\text{Cl}$.
102. Read the following 14 statements on enantiomers, diastereomers and physical properties and find how many of them are correct.
- (1) Enantiomers have identical melting points, boiling points and solubility in an achiral solvent.
 - (2) Diastereomers have different physical properties and can be separated by physical methods such as fractional crystallisation.
 - (3) Enantiomers react at different rates with a chiral, non-racemic reagent.
 - (4) A pair of enantiomers can be separated by simple fractional distillation.
 - (5) cis-But-2-ene and trans-but-2-ene are diastereomers.
 - (6) meso-Tartaric acid and (+)-tartaric acid are enantiomers.
 - (7) Stereoisomers that are not mirror images of each other are called diastereomers.
 - (8) A compound with n different stereocentres can have a maximum of $2^n - 1$ pairs of enantiomers.
 - (9) Epimers differ in configuration at only one stereocentre and are diastereomers.
 - (10) Anomers are enantiomers of each other.
 - (11) Conformers of an alkane at room temperature are normally separable as individual stereoisomers.
 - (12) Constitutional isomers have the same molecular formula but a different connectivity of atoms.
 - (13) A pair of enantiomers rotates plane-polarised light through the same angle in the same direction.
 - (14) A racemic mixture and the corresponding pure enantiomer always have the same melting point.
103. Read the following 14 statements on specific rotation, racemisation and resolution and find how many of them are correct.
- (1) Specific rotation is given by $[\alpha] = \alpha / (l \cdot c)$ with l in dm and c in g mL^{-1} .
 - (2) For a dilute solution the observed rotation is directly proportional to concentration and tube length.
 - (3) A racemic mixture can be resolved into its enantiomers.
 - (4) A meso compound can be resolved into optically active components.
 - (5) Optical purity = $([\alpha] \text{ of mixture} / [\alpha] \text{ of pure enantiomer}) \times 100$.
 - (6) A mixture of 75% R and 25% S has an enantiomeric excess of 50%.

- (7) A sample whose $[\alpha]$ is exactly half that of the pure (+)-form contains 75% of the (+)-enantiomer.
 - (8) The optical rotation of a sample is independent of the wavelength of light used.
 - (9) Specific rotations are conventionally quoted for the sodium D line (589 nm), written $[\alpha]_D$.
 - (10) Resolution through diastereomeric salts requires an optically pure resolving agent.
 - (11) Chiral amines with three different groups on nitrogen are readily resolved at room temperature.
 - (12) The sign of rotation can be predicted from the R/S descriptor.
 - (13) The observed rotation changes sign when the polarimeter tube is turned end for end.
 - (14) Mutarotation is the change in optical rotation of a freshly prepared sugar solution to an equilibrium value.
- 104.** Read the following 14 statements on special cases and counting of stereoisomers and find how many of them are correct.
- (1) An allene with two different substituents on each terminal carbon is chiral.
 - (2) Buta-1,2-diene ($\text{CH}_2=\text{C}=\text{CH}-\text{CH}_3$) is optically active.
 - (3) 6,6'-Dinitrodiphenic acid is optically active because rotation about the biaryl bond is restricted.
 - (4) trans-Cyclooctene is chiral and can be resolved.
 - (5) Spiro[3.3]heptane-2,6-dicarboxylic acid is chiral although it has no stereogenic carbon.
 - (6) Hexa-2,4-diene has four geometrical isomers.
 - (7) 1,2-Dichlorocyclopropane has three stereoisomers.
 - (8) 1,3-Dichlorocyclobutane is optically active.
 - (9) 2-Chloro-3-methylpentane has four stereoisomers.
 - (10) Menthol (2-isopropyl-5-methylcyclohexanol) has eight stereoisomers.
 - (11) Glycerol (propane-1,2,3-triol) is optically active.
 - (12) Cyclohexane-1,2,3-triol exists as four stereoisomers.
 - (13) $\text{CH}_3\text{CH}=\text{CH}-\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$ has four stereoisomers.
 - (14) Methylcyclohexane has two stereoisomers.

Part M — Statements (mixed topics)

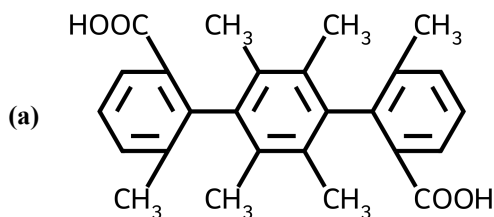
- 105.** Read the following 14 statements on pairs of stereoisomers, enantiomers and diastereomers and find how many of them are correct.
- (1) A pair of stereoisomers can be configurational isomers.
 - (2) A pair of stereoisomers can be diastereomers.
 - (3) A pair of stereoisomers can be constitutional isomers.
 - (4) A pair of stereoisomers can be tautomers.
 - (5) A pair of conformers (e.g. staggered and eclipsed ethane) is a pair of stereoisomers.
 - (6) A pair of stereoisomers can be enantiomers.
 - (7) A pair of stereoisomers can be positional isomers.
 - (8) Stereoisomers that are not mirror images of each other are diastereomers.
 - (9) In every case a pair of enantiomers has a mirror-image relationship.
 - (10) If a compound has an enantiomer it must be chiral.
 - (11) If a compound has a diastereomer it must be chiral.
 - (12) Any molecule containing a single asymmetric carbon (and no other stereo-element) must be chiral.
 - (13) Any chiral compound with a single asymmetric carbon must have a positive rotation if it has the R configuration.
 - (14) cis-trans (E/Z) isomers are diastereomers.
- 106.** Read the following 14 statements on meso compounds, D/L configuration and sign of rotation and find how many of them are correct.
- (1) A meso compound has chiral centres but exhibits no optical activity.
 - (2) A meso compound has no chiral centres and is therefore optically inactive.
 - (3) A meso compound is superimposable on its mirror image although it contains chiral centres.
 - (4) A meso compound is inactive because rotation caused by one half of the molecule is cancelled by the other half.
 - (5) A meso compound is inactive because the rotation of one molecule is cancelled by that of another molecule which is its mirror image.
 - (6) A compound with D configuration is always dextrorotatory.
 - (7) A compound with D configuration may be dextrorotatory or levorotatory.

- (8) A compound with L configuration is always levorotatory.
 - (9) A compound with L configuration may be dextrorotatory or levorotatory.
 - (10) D-(–)-Fructose has D configuration although it is levorotatory.
 - (11) D-(+)-Glucose is dextrorotatory.
 - (12) (S)-Lactic acid is the L-form and is dextrorotatory.
 - (13) In meso-2,3-dibromobutane one stereocentre is (R) and the other is (S).
 - (14) A meso compound can have only two stereocentres.
107. Read the following 14 statements on stereogenic atoms other than carbon, spiranes, biaryls and terphenyls and find how many of them are correct.
- (1) Methyl p-tolyl sulfoxide is optically active and can be resolved.
 - (2) Dimethyl sulfoxide is chiral.
 - (3) A neutral amine with three different groups on nitrogen can be resolved at room temperature.
 - (4) A quaternary ammonium ion with four different groups is optically active.
 - (5) Tröger's base is optically active because both bridgehead nitrogens are configurationally fixed.
 - (6) A phosphine with three different groups on phosphorus is configurationally stable at room temperature.
 - (7) In an amine oxide $R_1R_2R_3N \rightarrow O$ with three different groups the nitrogen is a stable stereocentre.
 - (8) Spiro[3.3]heptane-2,6-dicarboxylic acid is optically active.
 - (9) 6,6'-Dinitrodiphenic acid is optically active.
 - (10) 2,2',6,6'-Tetramethylbiphenyl is optically active.
 - (11) A terphenyl with two stereogenic axes and identical ends has three stereoisomers.
 - (12) Hexahelicene is chiral because of its helical shape.
 - (13) A molecule must contain a stereogenic centre to be optically active.
 - (14) Propadiene ($CH_2=C=CH_2$) is optically active.
108. Read the following 14 statements on projection formulas (Fischer, Newman, sawhorse, wedge) and find how many of them are correct.
- (1) A Fischer projection can be turned through 180° in the plane of the paper without changing the configuration.
 - (2) Rotating a Fischer projection by 90° in the plane changes it into the projection of the enantiomer.
 - (3) Interchanging two groups of a stereocentre in a Fischer projection retains the configuration.
 - (4) Two successive interchanges of groups in a Fischer projection retain the configuration.
 - (5) A Fischer projection may be lifted out of the plane and turned over.
 - (6) In a Fischer projection vertical bonds go away from the viewer and horizontal bonds come toward the viewer.
 - (7) In a Newman projection the bonds of the front carbon meet at the centre of the circle.
 - (8) Staggered and eclipsed Newman projections of the same molecule represent different configurations.
 - (9) A sawhorse projection and a Newman projection of the same conformer are equivalent descriptions.
 - (10) A Fischer projection represents the molecule in its staggered conformation.
 - (11) In a wedge-dash formula the solid wedge points toward the viewer and the dashed bond away from the viewer.
 - (12) Changing a single wedge bond at a stereocentre into a dashed bond inverts the configuration at that centre.
 - (13) The R/S descriptor of a stereocentre changes when only the projection formula used to draw it is changed.
 - (14) A Fischer projection correctly depicts the three-dimensional shape of ring compounds directly.
109. Read the following 14 statements on resolution, racemic modifications and optical purity and find how many of them are correct.
- (1) A racemic modification can be a racemic compound, a conglomerate or a solid solution.
 - (2) A conglomerate can be resolved by mechanical separation of the two kinds of crystals.
 - (3) A racemic acid treated with an optically pure base gives two diastereomeric salts of different solubility.
 - (4) After separating the diastereomeric salts, treatment with strong acid liberates the enantiomers.
 - (5) Enantiomeric excess equals optical purity when the rotation is linear in composition.
 - (6) A mixture of 90% (+) and 10% (–) enantiomer has an optical purity of 90%.
 - (7) Chromatography on a chiral stationary phase can separate enantiomers.
 - (8) A sample of 50% enantiomeric excess shows half the specific rotation of the pure enantiomer.
 - (9) A racemic mixture may show a small non-zero rotation.
 - (10) Enantiomers have different magnitudes of specific rotation at the same wavelength and temperature.

- (11) Diastereomers generally have unrelated specific rotations.
- (12) A pair of diastereomers can both be optically active.
- (13) Enzymes can discriminate between enantiomers, which forms the basis of biochemical resolution.
- (14) A solution containing equal masses of (+)-tartaric acid and meso-tartaric acid is optically inactive.
110. Read the following 14 statements on cyclic and bridged compounds and find how many of them are correct.
- (1) cis-1,2-Dichlorocyclopentane is a meso compound.
 - (2) trans-1,2-Dichlorocyclopentane exists as a pair of enantiomers.
 - (3) cis-1,4-Dimethylcyclohexane is optically active.
 - (4) trans-1,3-Dimethylcyclohexane is chiral.
 - (5) cis-1,3-Dimethylcyclohexane is a meso compound.
 - (6) Ring flipping converts one enantiomer of trans-1,2-dimethylcyclohexane into its mirror image.
 - (7) Chair conformers of a chiral cyclohexane derivative cannot be separated at room temperature.
 - (8) 1,2-Dichlorocyclopropane has three stereoisomers.
 - (9) Norbornane has bridgehead carbons but is achiral.
 - (10) cis- and trans-Decalin are conformational isomers.
 - (11) Cyclohexane-1,2,3-triol exists as two meso forms and one pair of enantiomers.
 - (12) trans-Cyclooctene is chiral.
 - (13) trans-Cyclohexene is stable and isolable at room temperature.
 - (14) cis- and trans-1,4-Dichlorocyclohexane are both optically inactive.
111. Read the following 14 statements on conformations and optical activity and find how many of them are correct.
- (1) gauche-Butane is a chiral conformation, but butane is optically inactive.
 - (2) anti-Butane has a centre of symmetry.
 - (3) The barrier to rotation about the C–C bond of ethane is about 12 kJ mol⁻¹.
 - (4) Gauche conformers of butane can be isolated at room temperature.
 - (5) Optical activity of a compound with rapidly interconverting chiral conformers is determined by its configuration.
 - (6) meso-Tartaric acid has chiral conformations but is optically inactive.
 - (7) Rotation about a C–C single bond can convert one enantiomer into the other.
 - (8) Biphenyls with bulky ortho substituents show atropisomerism.
 - (9) Atropisomers are conformers that can be isolated because rotation is hindered.
 - (10) Staggered ethane is more stable than eclipsed ethane mainly because of torsional strain in the eclipsed form.
 - (11) In n-butane the gauche conformer is more stable than the anti conformer.
 - (12) gauche conformers of ethane-1,2-diol are stabilised by intramolecular hydrogen bonding.
 - (13) 1,2-Difluoroethane prefers the gauche conformation (gauche effect).
 - (14) In eclipsed ethane every H–C–C–H dihedral angle is 0°.
112. 'A pair of stereoisomers might be classified in various ways depending on their exact nature.' Read the following 14 statements and find how many of them could correctly describe a pair of stereoisomers (assuming their nature permits).
- (1) They might be meso isomers.
 - (2) They might be tautomers.
 - (3) They might be enantiomers.
 - (4) They might be diastereomers.
 - (5) They might be conformational isomers.
 - (6) They might be constitutional isomers.
 - (7) They might be configurational isomers.
 - (8) They might be geometrical isomers.
 - (9) They might be optical isomers.
 - (10) They might be rotamers.
 - (11) They might be positional isomers.
 - (12) They might be functional isomers.
 - (13) They might be homomers.
 - (14) They might be chain isomers.

Part N — Passage-based questions

113. PASSAGE. In ortho-substituted biaryls, rotation about the aryl–aryl bond is hindered when each ring carries two different ortho groups, so the bond becomes a stereogenic axis. In the terphenyl (a) shown, the central ring carries four methyl groups and each outer ring carries the ortho groups COOH and CH₃. Question 1: How many stereogenic axes does (a) contain?



114. (Passage, Question 2) How many stereoisomers are possible for the terphenyl (a)?
115. (Passage, Question 3) How many of the stereoisomers of (a) are optically active?
116. (Passage, Question 4) How many of the stereoisomers of (a) are meso (achiral)?
117. (Passage, Question 5) How many pairs of enantiomers exist among the stereoisomers of (a)?
118. (Passage, Question 6) If one outer ring of (a) is changed to carry the ortho groups NO₂ and CH₃ instead (the other stays COOH/CH₃), how many stereoisomers are possible?

ANSWER KEY

1.	24	25.	6	49.	14	73.	8	97.	-4.5
2.	3	26.	7	50.	8	74.	48	98.	18
3.	5	27.	7	51.	4	75.	13.3	99.	8
4.	8	28.	6	52.	4	76.	48	100.	10
5.	10	29.	8	53.	5	77.	36	101.	9
6.	5	30.	6	54.	7	78.	8	102.	8
7.	4	31.	5	55.	4	79.	-52	103.	9
8.	4	32.	3544	56.	11	80.	80	104.	9
9.	4	33.	6	57.	19	81.	2	105.	9
10.	16	34.	6	58.	10	82.	52.48	106.	9
11.	4	35.	4	59.	2	83.	10	107.	9
12.	6	36.	1	60.	12	84.	13.5	108.	8
13.	3	37.	7	61.	5	85.	25	109.	10
14.	8	38.	8	62.	9	86.	30	110.	10
15.	4	39.	7	63.	8	87.	50	111.	10
16.	4	40.	4	64.	32	88.	80	112.	8
17.	2	41.	6	65.	4	89.	28	113.	2
18.	9	42.	5	66.	27	90.	16	114.	3
19.	10	43.	7	67.	20	91.	8	115.	2
20.	13	44.	6	68.	16	92.	7	116.	1
21.	9	45.	4	69.	2	93.	13	117.	1
22.	9	46.	20	70.	18	94.	16	118.	4
23.	9	47.	22	71.	6	95.	6		
24.	7	48.	28	72.	104	96.	10		